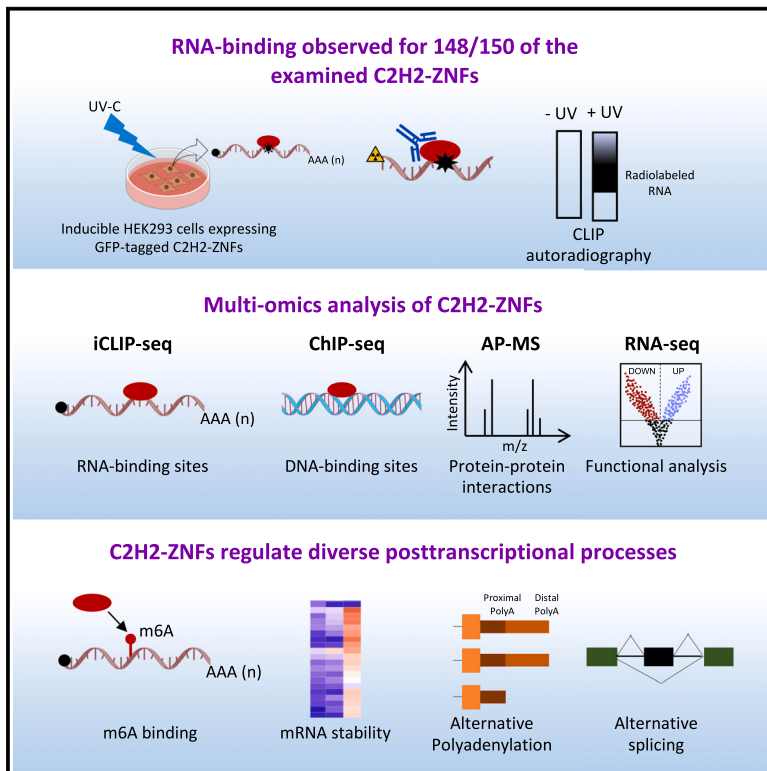


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C2H2-zinc-finger transcription factors bind RNA and function in diverse post-transcriptional regulatory processes

Graphical abstract



Authors

Syed Nabeel-Shah, Shuye Pu, James D. Burns, ..., Timothy R. Hughes, Benjamin J. Blencowe, Jack F. Greenblatt

Correspondence

jack.greenblatt@utoronto.ca

In brief

Nabeel-Shah et al. report that transcription factors of the C2H2-ZNF family also function as RNA-binding proteins that participate in multiple post-transcriptional processes, including alternative splicing, alternative polyadenylation, and m⁶A regulation. Integration of C2H2-ZNF iCLIP-seq data with other omics data reveals multiple modes of RNA binding by C2H2-ZNFs and uncovers mechanistic details.

Highlights

- 148 of the 150 examined C2H2-ZNFs bind directly to RNA in cells
- Denaturing IP conditions do not abolish C2H2-ZNFs' RNA binding
- iCLIP-seq for 94 RBPs, including 51 C2H2-ZNFs, reveals distinct binding preferences
- C2H2-ZNFs function in RNA-related processes ranging from splicing to m⁶A regulation

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Resource

C2H2-zinc-finger transcription factors bind RNA and function in diverse post-transcriptional regulatory processes

Syed Nabeel-Shah,^{1,2} Shuye Pu,¹ James D. Burns,^{1,2} Ulrich Braunschweig,¹ Nujhat Ahmed,^{1,2} Giovanni L. Burke,^{1,2} Hyunmin Lee,^{1,3} Ernest Radovani,¹ Guoqing Zhong,¹ Hua Tang,¹ Edyta Marcon,^{1,4} Zhaolei Zhang,^{1,3} Timothy R. Hughes,^{1,2} Benjamin J. Blencowe,^{1,2} and Jack F. Greenblatt^{1,2,5,*}

¹Donnelly Centre, University of Toronto, Toronto, ON M5S 3E1, Canada

²Department of Molecular Genetics, University of Toronto, Toronto, ON M5S 1A8, Canada

³Department of Computer Sciences, University of Toronto, Toronto, ON M5S 1A8, Canada

⁴Department of Laboratory Medicine and Pathobiology, University of Toronto, Toronto, ON M5S 1A8, Canada

⁵Lead contact

*Correspondence: jack.greenblatt@utoronto.ca

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SUMMARY

Cys2-His2 zinc-finger proteins (C2H2-ZNFs) constitute the largest class of DNA-binding transcription factors (TFs) yet remain largely uncharacterized. Although certain family members, e.g., GTF3A, have been shown to bind both DNA and RNA, the extent to which C2H2-ZNFs interact with—and regulate—RNA-associated processes is not known. Using UV crosslinking and immunoprecipitation (CLIP), we observe that 148 of 150 analyzed C2H2-ZNFs bind directly to RNA in human cells. By integrating CLIP sequencing (CLIP-seq) RNA-binding maps for 50 of these C2H2-ZNFs with data from chromatin immunoprecipitation sequencing (ChIP-seq), protein-protein interaction assays, and transcriptome profiling experiments, we observe that the RNA-binding profiles of C2H2-ZNFs are generally distinct from their DNA-binding preferences and that they regulate a variety of post-transcriptional processes, including pre-mRNA splicing, cleavage and polyadenylation, and m⁶A modification of mRNA. Our results thus define a substantially expanded repertoire of C2H2-ZNFs that bind RNA and provide an important resource for elucidating post-transcriptional regulatory programs.

INTRODUCTION

Post-transcriptional gene regulation plays essential roles throughout development. RNA-binding proteins (RBPs) are thought to regulate essentially all aspects of RNA metabolism¹ and many participate in multiple post-transcriptional processes. For example, individual Ser/Arg-repeat (SR) proteins and heterogeneous ribonucleoproteins (hnRNPs) have been shown to regulate alternative splicing (AS), alternative cleavage and polyadenylation (APA), mRNA transport, and translation.^{1–4} Reflecting their extensive roles in gene regulation, RBPs have been increasingly linked to human pathologies, including brain disorders and cancers.⁵

Human Cys2-His2 zinc-finger proteins (C2H2-ZNFs), the largest class of putative transcription factors (TFs), with more than 700 annotated members, are characterized by tandem arrays of ZNF domains (~10 on average).⁶ They often also possess one or more effector domains, including BTB/POZ (BR-C, ttk, and bab/Pox virus and zinc finger), KRAB (Krüppel-associated box), and/or SCAN (SRE-ZBP, CTFin51, AW-1, and number 18 cDNA). The roughly half of C2H2-

ZNFs containing KRAB domains mostly bind specific types of endogenous retroelements (EREs) and are thought to repress transcription by interacting with TRIM28 (tripartite motif-containing 28; also called KAP1), which recruits the transcriptional repressor SETDB1, a histone H3 lysine 9 (H3K9) trimethylase.⁷

We previously showed that >60% of C2H2-ZNFs bind specific DNA motifs in cells⁸ and that C2H2-ZNFs interact with proteins involved in diverse biological functions, including various aspects of transcriptional and post-transcriptional regulation.⁸ Although widely accepted as DNA-binding transcription factors (TFs), certain C2H2-ZNFs (e.g., general TFIIIA [GTF3A], CCCTC-binding factor [CTCF], Wilms' tumor 1 [WT1], and Yin-Yang 1 [YY1]) have been shown to regulate RNA-dependent processes,^{9–11} suggesting that C2H2-ZNFs might have wider regulatory roles. However, the prevalence, diversity, and functional significance of C2H2-ZNF-RNA interactions have remained largely unexplored. Here, we utilized genome-wide and biochemical approaches to interrogate the RNA-binding capacity and RNA-associated functions of the C2H2-ZNF family of TFs.

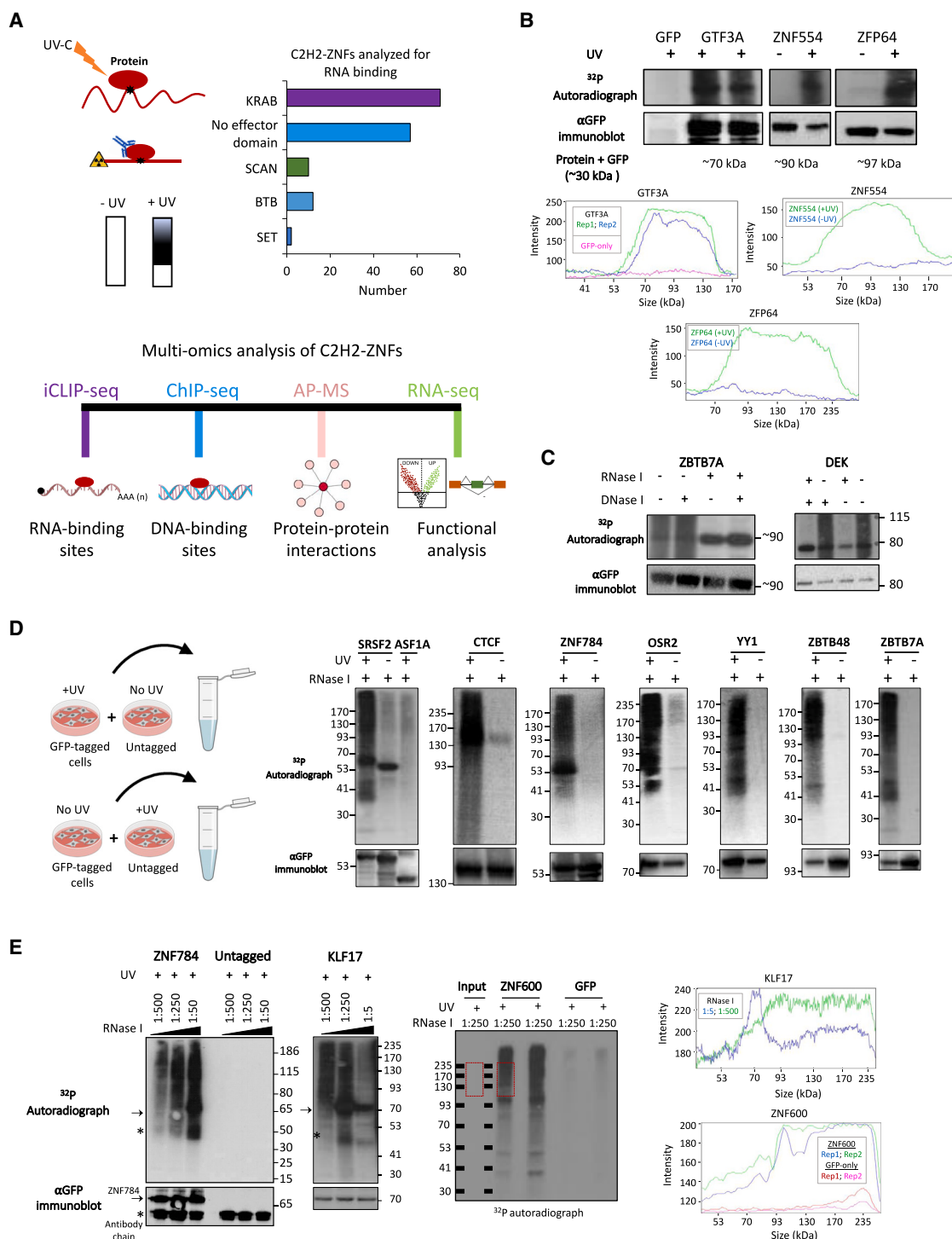


Figure 1. C2H2-ZNFs crosslink to RNA in cells

(A) Left, schematic representation of CLIP-autoradiography procedure. Right, bar graph shows the number of C2H2-ZNFs with various effector domains that were analyzed by CLIP autoradiography. Bottom, schematic of various assays that were used to characterize C2H2-ZNFs.

(B) Top, representative results of CLIP autoradiography. The presence of radioactive smearing is considered as evidence for RNA binding. Immunoblots show the recovery of the tagged proteins. Bottom, quantification of ³²P RNA intensities for +UV and –UV samples, measured by ImageJ (y axis). Quantification was performed across the shown lengths of each gel (x axis), which are scaled to predicted molecular weights.

(C) Autoradiographs of ³²P-labeled protein-RNA complexes after RNase I and/or DNase I over-digestion. The bottom panel western blots indicate the recoveries of the tagged proteins.

(legend continued on next page)

RESULTS

RNA binding by C2H2-ZNFs

To systematically interrogate their RNA-binding capacity, we utilized inducible HEK293 cell lines expressing 150 human C2H2-ZNFs fused to EGFP, including 71, 12, 10, and 57 with KRAB, BTB, SCAN, and no known effector domains, respectively (Figure 1A; Table S1). By using crosslinking and immunoprecipitation (CLIP), followed by radio-labeling of the RNA, SDS-PAGE, and autoradiography, we found that 98% (148/150) crosslinked robustly to RNA (Figures 1A, 1B, S1A, and S1B), resulting in radioactive signals with apparent molecular weights greater than those of the tagged proteins. Previously characterized RBPs, including GTF3A (Figure 1B) and splicing factors SRSF1 and MBNL2 (Figure S1C), also crosslinked readily to RNA, whereas, importantly, the EGFP negative control (Figure 1B) and two C2H2-ZNFs, ZNF33A and SP4 (Figure S1B), yielded no observable radioactive signals. With only several exceptions, little RNA association was observed in the absence of UV crosslinking, indicating that radioactive signals were generally dependent on the formation of UV-induced RNA-protein complexes (Figures 1B, S1A, and S1B).

For comparison purposes, we also analyzed the TFs MYC, MYB, and GATA1, which are not C2H2-ZNFs, and the chromatin-associated proteins TRIM28,¹² DEK, HMGB2, H1FO, RPRD1A, RPRD1B, and RPRD2. TRIM28 and the other TFs and chromatin-associated proteins crosslinked robustly to RNA (Figure S1D), whereas two histone chaperones, anti-silencing factor 1 (ASF1) and histone cell cycle regulator (HIRA), which are not known to bind RNA, did not exhibit RNA binding (Figure S1B).

Certain proteins (e.g., ZNF513, ZBTB6, and ZBTB26) became radiolabeled in the non-UV samples, presumably because of protein phosphorylation (Figure S1A). Such protein phosphorylation has often been observed in CLIP experiments for certain well-characterized RBPs, e.g., SRSF1, and is attributed to endogenous protein kinases that co-precipitated with the immunoprecipitated proteins.¹³ Importantly, radiolabeled smearing, indicative of higher-molecular-weight protein-RNA complexes, was not observed with non-UV controls (Figures S1A and S1B). Some high-molecular-weight proteins (e.g., ZNF594, ZFP64, and ZNF341) exhibited degradation in the initial CLIP experiments (Figure S1E), which we found can often be mitigated by increasing the volume of lysis buffer. Strongly crosslinked proteins almost always exhibited a relatively weaker band on immunoblots compared with non-UV controls (e.g., ZNF281, ZNF610, ZNF16, and ZBTB2) (Figure S1A). When we utilized higher RNase 1 concentration (1:100) in CLIP experiments, protein detection was improved on immunoblots (Figure S1F), suggesting that crosslinked RNA might hinder

GFP antibody binding to target proteins on the membrane. Consistent with this possibility, treating UV-crosslinked whole-cell lysates with RNase I substantially enhanced protein detection on immunoblots, such that the signals became comparable to those of non-UV control samples (Figure S1G).

To assess whether RBP overexpression significantly affects their RNA associations, we compared the RNA binding of tagged, overexpressed *CTCF*, *DEK*, and *SRSF1* with their counterparts endogenously tagged with C-terminal EGFP using the CRISPR-Cas9 system (Figure S1H). Endogenous and overexpressed proteins showed comparable RNA binding in CLIP-autoradiography experiments (Figure S1H), suggesting that RBP overexpression only marginally affects their RNA-binding capacity (also see below). Consistently, we previously showed that endogenous and overexpressed SP1, a C2H2-ZNF protein, robustly crosslink to RNA in cells.¹⁴ These results also agree with previous CLIP sequencing (CLIP-seq) studies showing that endogenous and overexpressed HNRNPA1 have similar RNA-binding profiles.¹⁵

The radiolabeled bands were indeed RNA dependent, rather than DNA dependent, because the radioactive signals from many immunoprecipitated RNA-protein complexes were diminished by over-digestion with RNase I, but not DNase I, supporting the conclusion that the C2H2-ZNFs ZNF784, ZNF610, ZBTB2, ZBTB6, ZBTB26, and ZBTB7A, as well as the chromatin-associated protein, DEK, bind directly to RNA (Figures 1C and S1I). Moreover, RNA over-digestion reduced the radioactive smears to sharp bands and led to increased recovery of the tagged proteins on immunoblots (Figure S1I), consistent with our above-described results.

Although our optimized individual-nucleotide-resolution CLIP (iCLIP) washes are highly stringent,¹⁶ greatly reducing the background of non-specifically co-purifying proteins compared with conventional immunoprecipitations (Figure S1J), it was still possible that RNA signals could arise from protein-RNA binding after cell lysis or interacting, crosslinked RBPs that persisted during the immunoprecipitation and SDS-PAGE procedures. To examine these possibilities, we first used FLAG-tagged proteins and urea to resolubilize protein-RNA complexes under denaturing conditions, then reprecipitated the proteins with anti-FLAG antibodies prior to autoradiography, finding that robust RNA signals persisted for all examined C2H2-ZNFs despite denaturing conditions that should have disrupted non-covalent protein-protein and protein-RNA interactions (Figure S1K).

To further rule out such spurious interactions, we mixed UV-treated cells containing tagged SRSF2, which regulates splicing, or six different tagged C2H2-ZNFs with untreated, untagged cells prior to cell lysis, or mixed UV-untreated cells expressing tagged proteins with UV-treated, untagged cells (Figure 1D).

(D) Left, schematic of mixed +UV and -UV cells used in CLIP autoradiography. Right, CLIP autoradiography of the representative proteins using mixed cells. (E) Left, two representative results for optimization of RNA fragmentation using various dilutions of RNase I. Top: CLIP autoradiography of ³²P-labeled protein-RNA complexes. Bottom: western blotting analysis to show the recoveries of the tagged proteins. Middle, representative images of autoradiographs of input and ³²P-labeled CLIP ZNF600-RNA complexes after RNase I digestion using optimal RNase I concentrations. Input material (2% of the starting material) was resolved alongside the IPs. Red boxes indicate the excised areas. Arrows indicate positions of tagged proteins. Asterisks indicate antibody chains. Right, quantification of ³²P RNA intensities for the indicated CLIP samples measured by ImageJ (y axis). Quantification was performed across the shown lengths of each gel (x axis), which is scaled to predicted molecular weights. Figure legends are provided (Rep, replicate). Related Figure S1 and Table S1.

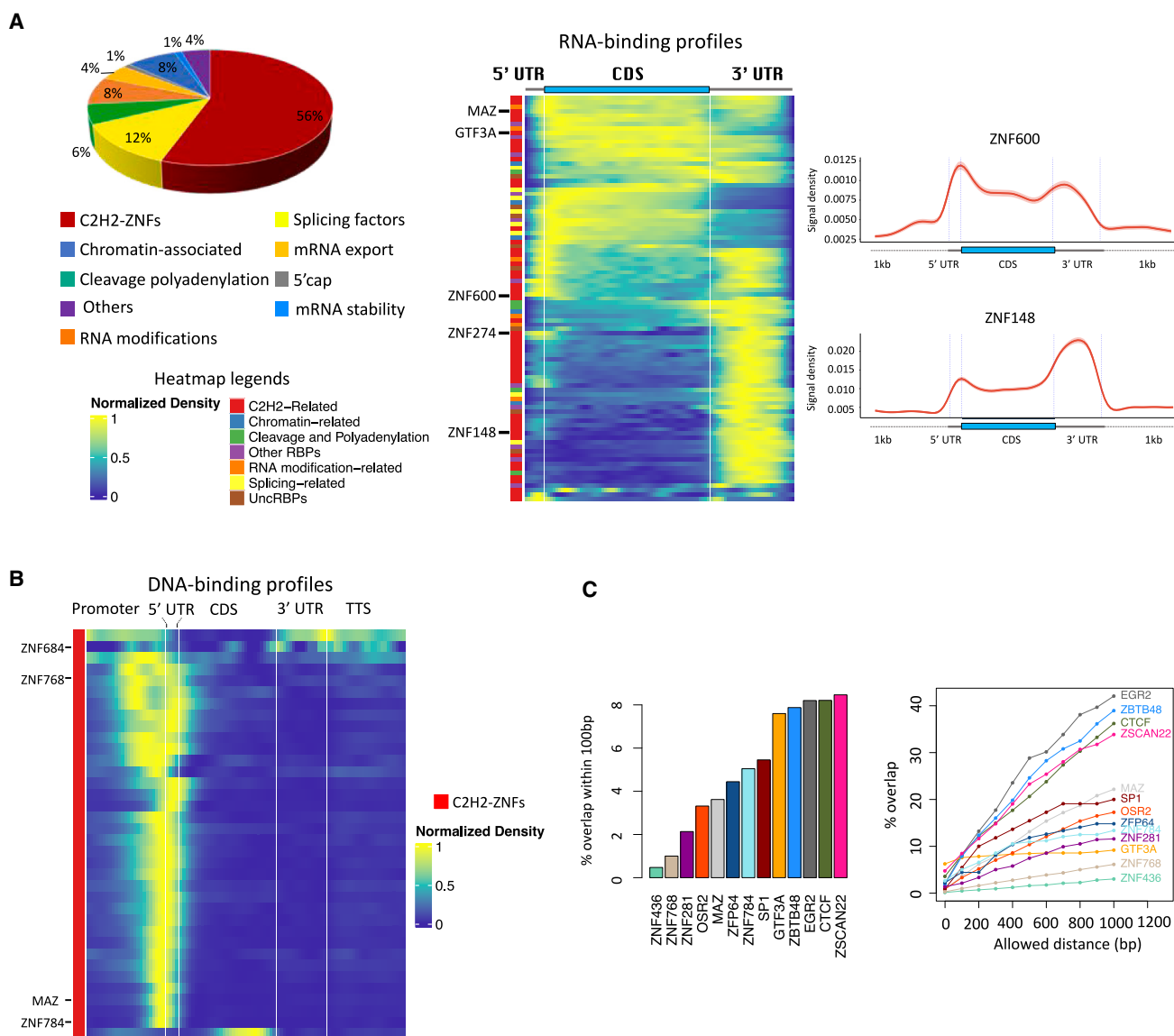


Figure 2. mRNA metagenome analyses indicate distinct RNA-binding patterns for C2H2-ZNFs

(A) Left, pie chart showing the distribution of the various proteins that were analyzed by iCLIP-seq. Middle, heatmap representation of iCLIP-seq profiles. Local enrichment across the mRNA was calculated using iCLIP-seq peaks and represented as a heatmap. Color indicates the average occupancy, normalized by setting the minimum value to zero (blue) and maximum to 1 (yellow) per row. Right, representative individual metagenome plots.

(B) Heatmap representation of metagenome profiles using ChIP-seq peaks for 36 C2H2-ZNFs. Normalized density is colored the same as for (A). Promoter: 1 kb upstream of TSS.

(C) Left, bar plot representation of % overlap of iCLIP peaks with ChIP-seq peaks (100 nt window around the peak center) for 13 representative C2H2-ZNFs. Right, line graph shows the % overlap of iCLIP peaks with ChIP peaks while increasing the allowed window size around the ChIP peaks.

Refer to [Tables S2](#) and [S3](#).

For all these proteins, including CTCF and YY1, radioactive signals were observed only if UV was used to treat the cells with the tagged protein, ruling out the possibility of protein-protein or protein-RNA interactions occurring after cell lysis. Moreover, no radiolabeled RNA signal was obtained for the negative control protein ASF1 in these experiments, indicating that crosslinking-related background was minimal in our experiments (Figure 1D). We conclude that RNA binding is a general property of C2H2-

ZNFs and also prevalent among other classes of human TFs and many chromatin-associated proteins.^{17–20}

Large-scale profiling of RNA-binding sites of C2H2-ZNFs and other RBPs

Given the unexpectedly widespread binding of C2H2-ZNFs to RNA, we used iCLIP-seq to map the binding sites of 51 C2H2-ZNFs expressed at varying levels in HEK293 cells (Figure S1L). For

control and comparison purposes, 11 RNA-binding/chromatin-associated proteins and 32 canonical RBPs with wide-ranging functions and motif preferences were analyzed in parallel (Figure 2A, left). To obtain optimal-size RNA fragments (50–300 nucleotides [nt]) for each protein and visualize the quality of immunoprecipitated RNAs, we performed CLIP autoradiography with varying RNase I concentrations, observing a clear RNase-dependent size shift of protein-RNA complexes for all examined proteins (Figures 1E and S1M) and reinforcing direct binding of C2H2-ZNFs to RNA in cells. After RNase I optimization and iCLIP-seq with at least 2 biological replicates (Figure S1N; Table S2), yielding 240 individual datasets, including EGFP-only-expressing controls, samples with <0.5 million uniquely mapping reads (e.g., CBLL1/HAKAI) were eliminated from further analyses, leaving 238 high-quality iCLIP-seq datasets representing 93 proteins.

We defined RNA-binding sites as “crosslinking-induced truncation sites” (CITSs) (false discovery rate [FDR] $\leq$ 0.01). Because EGFP libraries were dominated by PCR duplicates and did not yield many specific binding peaks, we normalized the iCLIP-seq datasets against 24 merged size-matched-input (SMI) control datasets generated in parallel for 23 proteins with molecular weights ranging from ~30 to 130 kDa (Table S2), using differential analysis based on read counts from regions corresponding with the identified CITS (see STAR Methods). Using a uniform experimental system, i.e., inducible EGFP-tagged proteins, combined with the breadth of our data, enabled identification and removal of ~8,000 crosslink hotspots (Table S3), yielding the remaining high-quality peaks used for downstream analyses. iCLIP-seq performed in cells expressing endogenously tagged SRSF1 and DEK strongly correlated (R : ~0.9) with data from cells where these proteins were overexpressed near endogenous levels (Figures S2A and S2B), further indicating that RBP overexpression does not substantially change their binding sites.

All proteins predominantly targeted protein-coding transcripts followed by unannotated intergenic transcripts (Figure S3). In absolute numbers, most peaks were within introns, suggesting that most C2H2-ZNFs likely bind nascent transcripts (Figure S4A). After normalizing to the length of various transcript features, various proteins showed enrichment in coding sequences (CDSs), 5' and 3' untranslated regions (UTRs), and/or introns (Figure S4B). Figures 2A and S5 illustrate RNA binding as heatmaps along the CDS and 5' and 3' UTRs of spliced mRNA. Clustering revealed binding patterns that could be broadly grouped as targeting: (1) the whole transcript, (2) the CDS, (3) the 5' UTR and CDS, (4) both 5' and 3' UTRs, and (5) 3' UTRs (Figure 2A, right). Importantly, RNA binding by previously characterized RBPs was consistent with their known functional roles (e.g., 3' UTR enrichment of cleavage and polyadenylation [CPA] complex subunits CPSF1 [CPSF160], CPSF2, CPSF5 [NUDT21], and CPSF7) (Figure S5). Although many C2H2-ZNFs showed enrichment in 3' UTRs, certain C2H2-ZNFs preferentially bound 5' UTRs (e.g., ZNF768 and ZNF600) and/or the CDS (e.g., ZNF436 and GTF3A) (Figures 2A and S5).

Relationship between RNA and DNA binding by C2H2-ZNFs

Because C2H2-ZNFs can function as TFs, we also examined their DNA-binding profiles using our published chromatin immu-

noprecipitation sequencing (ChIP-seq) data.^{8,21} Except for ZNF384, which binds DNA in both 3' UTRs and promoters, and ZNF274, which binds coding region DNA, C2H2-ZNFs were preferentially enriched on promoter region DNA (and 5' UTRs) (compare Figures 2A, 2B, S5, and S6A). The overlap was small, <10% of iCLIP peaks overlapping ChIP-seq peaks extended 50 nt on each side (Figures 2C and S6B), suggesting that C2H2-ZNFs generally bind distinct RNA and DNA sequences. Increasing the window size for overlaps to 1 kb resulted in corresponding increases in overlap between iCLIP peaks and ChIP peaks for most C2H2-ZNFs, as expected for no correlation, with the notable exception of GTF3A (Figure 2C). The overlapping GTF3A iCLIP-seq peaks were concentrated in 5S genes and pseudogenes, consistent with demonstrations that GTF3A binds both 5S gene promoters and the 5S transcript.²² To further compare iCLIP- and ChIP-binding profiles, we generated an artificial set of ChIP-binding sites and constructed three prototype profiles to compare it with (see STAR Methods), finding that most C2H2-ZNFs followed curves reminiscent of random samples, again indicating that there is no spatial relationship between iCLIP peaks and ChIP peaks (Figure S7). Therefore, C2H2-ZNFs exhibit distinct RNA-binding patterns across various mRNA regions, and their RNA- and DNA-binding sites are generally different.

Although C2H2-ZNFs have distinct RNA- and DNA-binding sites, RNA binding might affect their chromatin association.²³ To examine this possibility, we utilized published quantitative proteomic data examining the effect of RNA on chromatin occupancy.²⁴ This revealed that inhibiting production of nascent RNAs with the RNA polymerase II transcription inhibitors triptolide (Trp) and flavopiridol (Fp) significantly increased the chromatin association of C2H2-ZNFs (Figure S8A). Conversely, canonical RBPs exhibited significantly reduced chromatin binding, as expected (Figure S8A). Similar results were obtained for cells treated with RNase A (Figure S8A). To validate these findings, we performed ChIP-qPCR in HEK293 cells treated with the transcription inhibitor actinomycin D, finding that PATZ1 and SP1 increased their chromatin occupancy at specific sites (Figure S8B). These observations indicate that RNA binding by C2H2-ZNFs usually inhibits their DNA binding.

C2H2-ZNFs bind to accessible sites within structured RNAs

To investigate the structural context of RNA binding by C2H2-ZNFs, we averaged *in vivo* click selective 2'-hydroxyl acylation and profiling experiment (icSHAPE) reactivity scores across their iCLIP peak summits, plus 50 nt on each side, by utilizing published icSHAPE data (+proteins) in HEK293 cells²⁵ or using purified RNAs (–proteins).²⁶ We also utilized RNAfold, which folds RNA sequences based on minimum free energy conformations and extracts the unpaired probability of each nucleotide,²⁷ yielding a sequence-based structural prediction. As expected, three single-stranded RBPs (ssRBPs), SRSF1, SRSF2, and ZRANB2, displayed central reactivity peaks, consistent with unpaired RNA, coinciding with their RNA crosslink sites (Figures S9A and S9B). Again, as expected, neither of two double-stranded RBPs (dsRBPs), ADAR1 and Staufen 2 (STAU-2), exhibited a central reactivity peak, using either our iCLIP-seq

data or previously reported enhanced CLIP (eCLIP)-seq data,²⁸ and the RNA appeared structured throughout the examined regions (Figures S9A and S9B).

Remarkably, except for WT1, CTCF, and perhaps OVOL2, and like SRSF1, SRSF2, and ZRANB2, most C2H2-ZNFs displayed central reactivity peaks around their crosslink sites, surrounded by RNA with low reactivity scores (i.e., more structured) (Figures S9A and S9B), for *in vivo* and *in vitro* icSHAPE datasets and *in silico* RNAfold predictions. Control datasets generated by random shuffling of sequences while maintaining their dinucleotide frequencies showed no central peaks in RNAfold analyses, as shown for Sp1 and ADAR1 (Figures S9A and S9B). WT1 was previously demonstrated to bind structured RNA,¹⁰ and our analysis suggests that CTCF might also bind dsRNA with stable structures. These results indicate that C2H2-ZNFs generally prefer crosslinking to RNA bases at sterically accessible nucleotides.

We then utilized the BEAM (BEAr Motifs finder) package to predict structures for interaction motifs in CLIP-seq data.²⁹ For the dsRBP STAU-2, eCLIP-seq data yielded a long hairpin motif predicted for at least 32% of STAU2-bound RNAs (Figure S10A). For the ssRBPs SRSF1, SRSF2, and ZRANB2, however, predicted structural motifs appeared artifactual, with large loops (11–18 nt) and very short stems (3–4 bp), likely representing single-stranded RNA (Figure S10A). Most C2H2-ZNFs preferred at least one short hairpin motif with a 4- to 5-bp stem and well-defined 5- to 8-nt loop present in 15%–30% of target RNAs (Figures S10A and S10B). As for SP1 (Figure S10A), predicted C2H2-ZNF structural motifs were enriched around iCLIP crosslink sites frequently occurring in predicted loop regions, similar to structural motifs targeted by WT1³⁰ and consistent with our icSHAPE/iCLIP analysis indicating that C2H2-ZNFs tend to crosslink in open regions of target RNAs surrounded by more structured RNA.

C2H2-ZNFs interact with RBPs involved in diverse post-transcriptional processes

Their ubiquitous RNA binding suggests that C2H2-ZNFs could potentially function in diverse post-transcriptional regulatory processes. Therefore, we utilized our published affinity purification-mass spectrometry (AP-MS) data to investigate their possible role(s) in RNA metabolism.⁸ Among the 24 C2H2-ZNFs for which we carried out iCLIP-seq experiments and AP-MS data were available,⁸ 17 had at least one significant RBP interaction partner (significance analysis of interactome [SAINT] score ≥ 0.93) (Figure 3A; Table S4). These interacting RBPs are functionally linked to post-transcriptional processes ranging from mRNA splicing (e.g., ZSCAN22) and *N*⁶-methyladenosine (m⁶A) regulation (e.g., KLF14) to translation (e.g., MYST2) (Figure 3A), CPA (e.g., SP1³¹), and mRNA export (ZNF684³²), suggesting that particular C2H2-ZNFs might function specifically in particular post-transcriptional regulatory processes.

Importantly, these protein-protein interactions of C2H2-ZNFs were not detected when AP-MS was performed under CLIP purification conditions using UV-crosslinked cell lysates from the iCLIP-seq experiments described above. For most proteins, no significant interacting protein was detected (FDR ≤ 0.01) (Table S5), indicating that RNA signals in CLIP-autoradiography

and iCLIP-seq experiments are not likely due to co-purifying RBPs.

Regulation of splicing by C2H2-ZNFs

To explore their potential roles in splicing regulation, we examined the RNA-binding profiles of C2H2-ZNFs around intron-exon junctions (Figures 3B and S11). Importantly, previously characterized splicing regulators like U2AF1, which recognizes 3' splice sites,^{33,34} SRSF1, which binds in exons, and TDP43 and MATR3, which target introns or distal intronic regions (>500 nt away from the splice site),^{35,36} exhibited their expected binding preferences (Figures 3B and S11).

Certain C2H2-ZNFs, like ZNF654 and ZNF677, exhibited enrichment immediately upstream of 3' splice sites, whereas others, such as ZNF600 and ZBTB7A, bound near 5' splice sites (Figures 3B and S11), suggesting that various C2H2-ZNFs might regulate splicing, perhaps by coordinating with other splicing regulators. For example, our AP-MS analysis indicated that ZNF768 reciprocally co-purified with SRSF1 (Figure 3C; Table S6), suggesting that ZNF768 might regulate splicing by associating with this factor. Analysis of our RNA sequencing (RNA-seq) data⁸ indicated that ZNF768 overexpression alters inclusion for >800 cassette exons (differential percent spliced in [dPSI] ≥ 10) (Figure S12A; Table S7). Among these exons, 244 and 425 had defined ZNF768- and SRSF1-binding sites (i.e., significant CITS [FDR ≤ 0.01] within the exons and ± 300 nt), respectively. Composite analysis indicated that exons with increased inclusion showed substantially more ZNF768 and SRSF1 binding than control exons (Figures S12B and S12C), whereas downregulated exons did not show such a trend and likely represent indirect effects. When we treated ZNF768-overexpressing cells with either siSRSF1 or control small interfering RNAs (siRNAs) (Figure S12D), ZNF768 overexpression increased inclusion of the examined target exons, and this increased inclusion was reversed upon concomitant SRSF1 depletion (Figure S12E). These data suggest that ZNF768 and SRSF1 may function cooperatively to regulate a subset of target exons.

Although most C2H2-ZNFs and other RBPs exhibited enriched binding in exons relative to introns, certain C2H2-ZNFs were enriched in introns downstream of 5' splice sites and in distal intronic regions (e.g., YY1, ZNF274, ZNF777, and ZNF467) (Figures 3B and S11). Among these intron-binders, ZNF467 is expressed very weakly, if at all, in HEK293 cells (Figure S13A). However, forced expression of ZNF467⁸ caused splicing defects in 353 cassette exons, with 314 exhibiting reduced inclusion (dPSI ≥ 10) (Figure S13B; Table S8). ZNF467 was significantly enriched in introns near the splice sites of downregulated exons compared with unaffected exons (Figure S13C), suggesting that intron binding by ZNF467 inhibits inclusion of certain target exons.

We then examined the oncoprotein ZBTB7A,³⁷ which showed mild intronic enrichment downstream of 5' splice sites (Figure 3B). In AP-MS experiments, we observed that ZBTB7A co-purifying proteins were highly enriched for splicing factors, including components of the U1 and U2 small nuclear ribonucleoprotein particle (snRNP) sub-complexes of the spliceosome (Figures S14A and S14B; Table S9). Interaction of ZBTB7A with U2 snRNP subunit SF3B1 was confirmed using

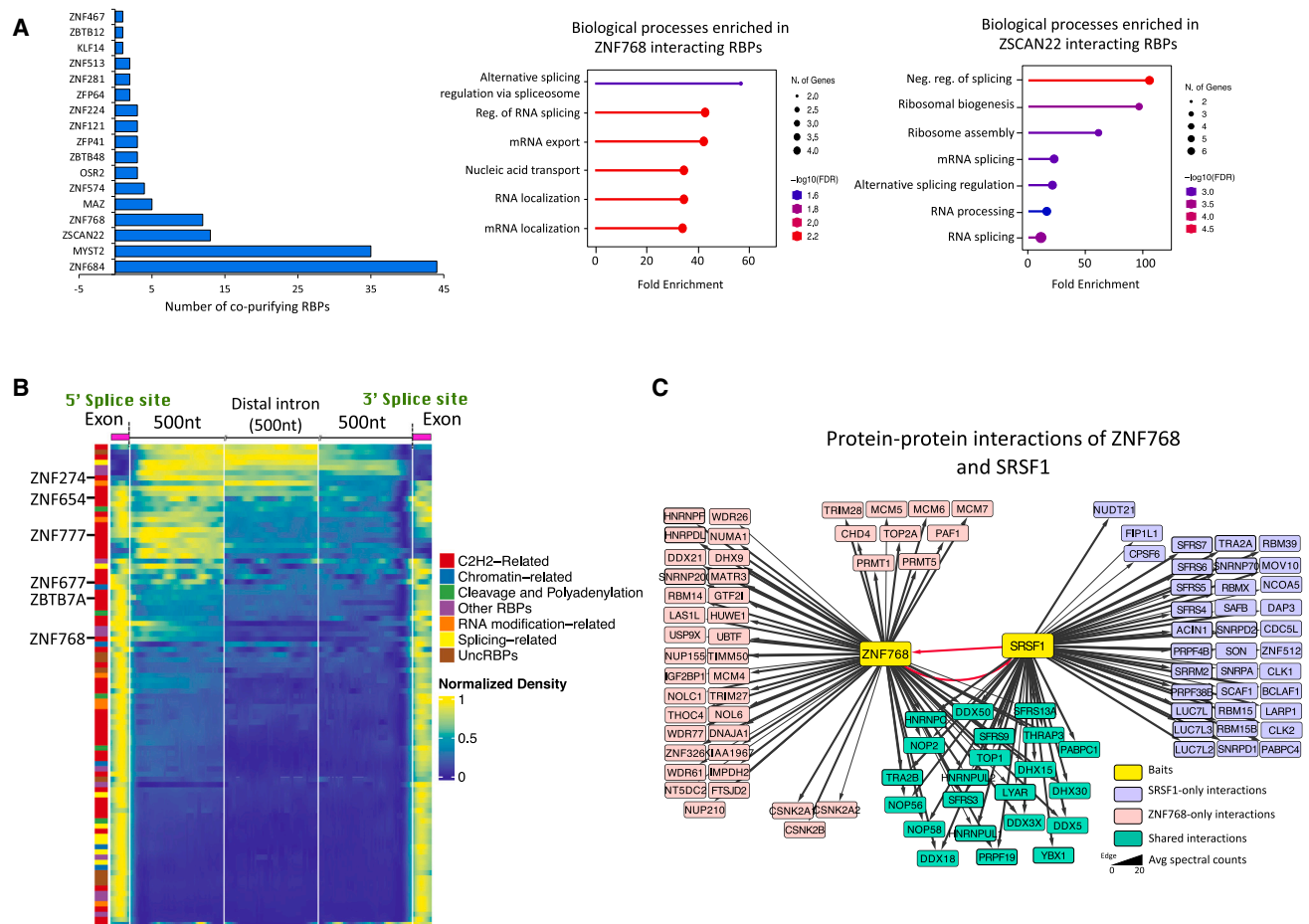


Figure 3. Some C2H2-ZNFs bind near splice sites and regulate alternative splicing

(A) Bar plot representation of number of RBPs that co-purify with the indicated C2H2-ZNFs (left). Gene Ontology (GO) enrichment analyses related to biological processes for the ZNF768 (middle) and ZSCAN22 (right) co-purifying RNA-binding proteins are shown. Refer to [Table S4](#).

(B) Heatmap representing the local enrichment of normalized iCLIP-seq signals within 100 and 500 nt of intron-exon junctions for exons and introns, respectively. Normalized density is colored the same as in [Figure 2](#).

(C) Protein-protein interaction networks of ZNF768 and SRSF1. AP-MS data were filtered against numerous (>40) control purifications, and prey proteins with Bayesian false discovery rate (BFDR) ≤ 0.01 are reported (see also [Tables S5](#) and [S6](#)).

co-immunoprecipitation (coIP) ([Figure 4A](#)). ZBTB7A knockdown followed by RNA-seq analysis revealed AS defects for only a few cassette exons (~60 events), without any clear trend in terms of down regulation or upregulation ([Figure 4B](#)), likely representing the indirect effects of altering expression of other splicing factors ([Figure S14C](#); [Tables S10](#) and [S11](#)). However, in ZBTB7A-depleted cells ([Figure 4B](#); [Table S10](#)), ~100 introns exhibited increased retention (differential percent intron retention [dPIR] ≥ 10) (versus only 24 with reduced retention) in genes significantly enriched in processes related to brain and central nervous system development ([Figure 4C](#)). We validated four of these by RT-PCR, utilizing RNA from independent ZBTB7A knockdowns (siZBTB7A # 2) ([Figure S14D](#)).

Consistently, ZBTB7A showed enriched binding near 5' and 3' splice sites in exons flanking affected introns, and iCLIP density was also significantly higher in affected introns (500 nt downstream and upstream of 5' and 3' splice sites, respectively) than in control introns ($p \leq 0.05$, [Figures 4D](#), [S15A](#), and [S15B](#)). Unlike

ZBTB7A, neither ZNF768 nor YTHDF2 exhibited any enrichment in siZBTB7A-regulated introns ([Figure S15C](#)). Moreover, the ZBTB7A RNA-binding signal was not enriched in introns affected by ZNF768 overexpression ([Figure S15D](#)), suggesting that RNA binding by ZBTB7A in target introns and/or flanking exons affects intron retention. Although intron lengths were not significantly different, ~35 introns with increased retention were first introns ([Figure S15E](#); [Table S10](#)). Although first introns in plants are thought to regulate expression,³⁸ genes with retained introns were rarely differentially expressed in ZBTB7A-depleted cells ([Figure S15F](#)). Although RNA binding supports a direct role in regulating target introns, the ZBTB7A ChIP-seq signal was also significantly enriched around the TSS of genes with introns exhibiting elevated retention ([Figure 4E](#)), suggesting potential co-transcriptional recruitment of ZBTB7A to pre-mRNAs ([Figure 4F](#)). In contrast, the ZNF768 ChIP-seq signal did not show any enrichment around siZBTB7A-regulated introns ([Figure S15G](#)). Although further studies are needed to determine precisely

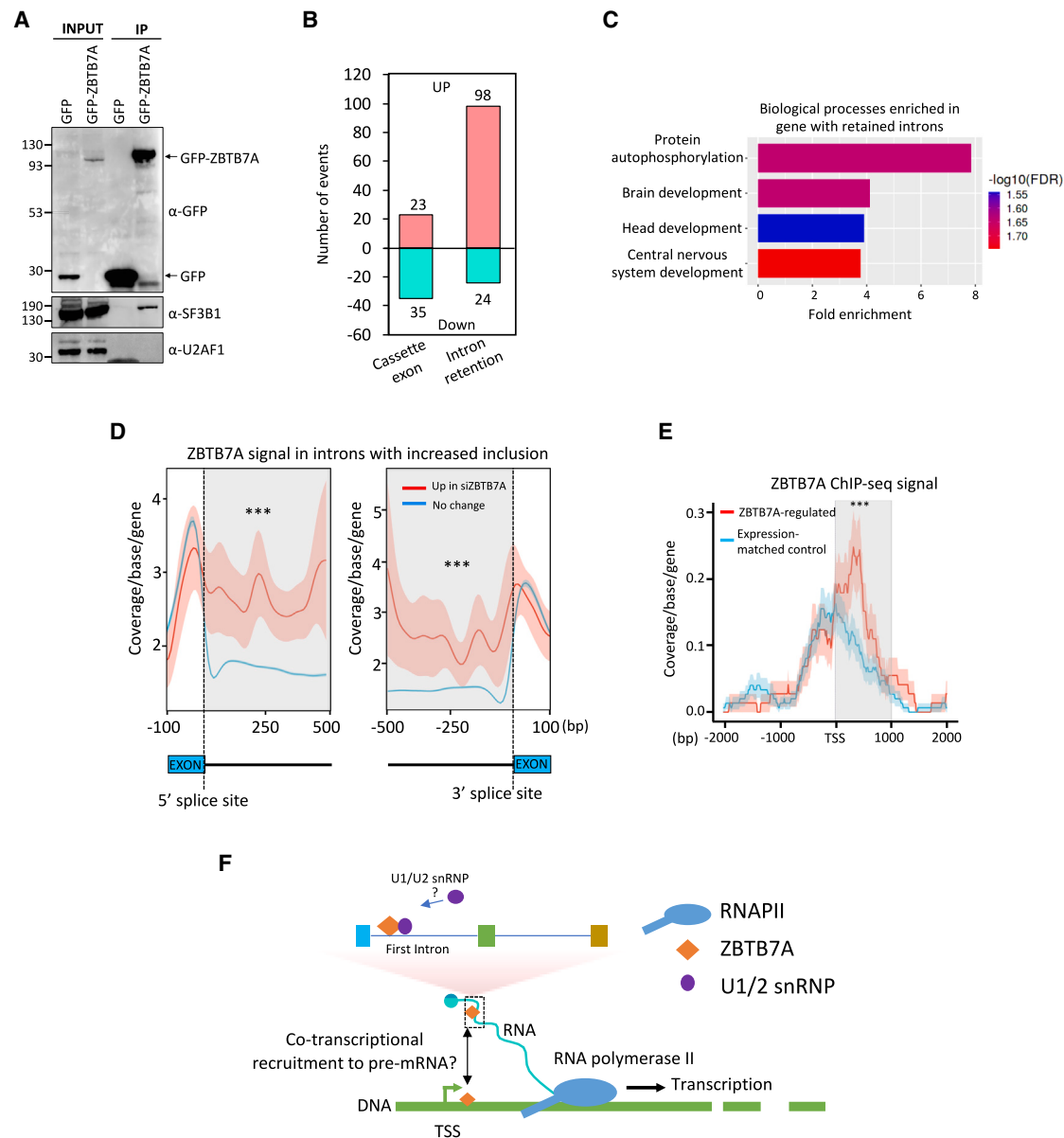


Figure 4. ZBTB7A modulates intron retention

(A) Co-immunoprecipitation (coIP) of ZBTB7A with SF3B1. Cell lysates prepared from GFP-ZBTB7A-expressing cells were treated with nuclease (Benzonase) prior to IPs with anti-GFP antibody. Blots were probed with the indicated antibodies.

(B) Bar plot showing the number of detected splicing changes for cassette exons ($dPSI \geq \pm 10$) and intron retention events ($dPIR \geq \pm 10$) in siZBTB7A RNA-seq data. Number of retained introns is significantly more than affected exons ($p = 1.531e-7$, Fisher's exact test).

(C) Bar plot showing GO enrichment analysis related to biological processes for genes whose introns exhibit increased retention in siZBTB7A cells.

(D) Plots of average iCLIP-seq signal in the introns and their flanking exons for introns whose inclusion levels are upregulated in siZBTB7A cells. Boxplots show the average iCLIP density in introns (500 nt downstream and upstream of 5' and 3' splice sites, respectively) whose retention levels either increase or do not change in siZBTB7A cells. Wilcoxon test, $***p \leq 0.001$. Control (no-change) introns were matched for length and percent intron retention (PIR) values.

(E) Average ChIP signal around TSS of those genes whose introns showed either increased retention or remained unchanged in siZBTB7A cells (control genes are expression matched; Wilcoxon test, $***p \leq 0.001$).

(F) Hypothetical model for ZBTB7A co-transcriptional recruitment to RNA and regulation of first introns.

how ZBTB7A and other C2H2-ZNFs regulate splicing, these examples indicate that C2H2-ZNFs might utilize distinct mechanisms to regulate splicing.

Role of C2H2-ZNFs in m⁶A regulation

Because m⁶A is the most prevalent internal modification of mRNAs, with broad roles in post-transcriptional processes,³¹

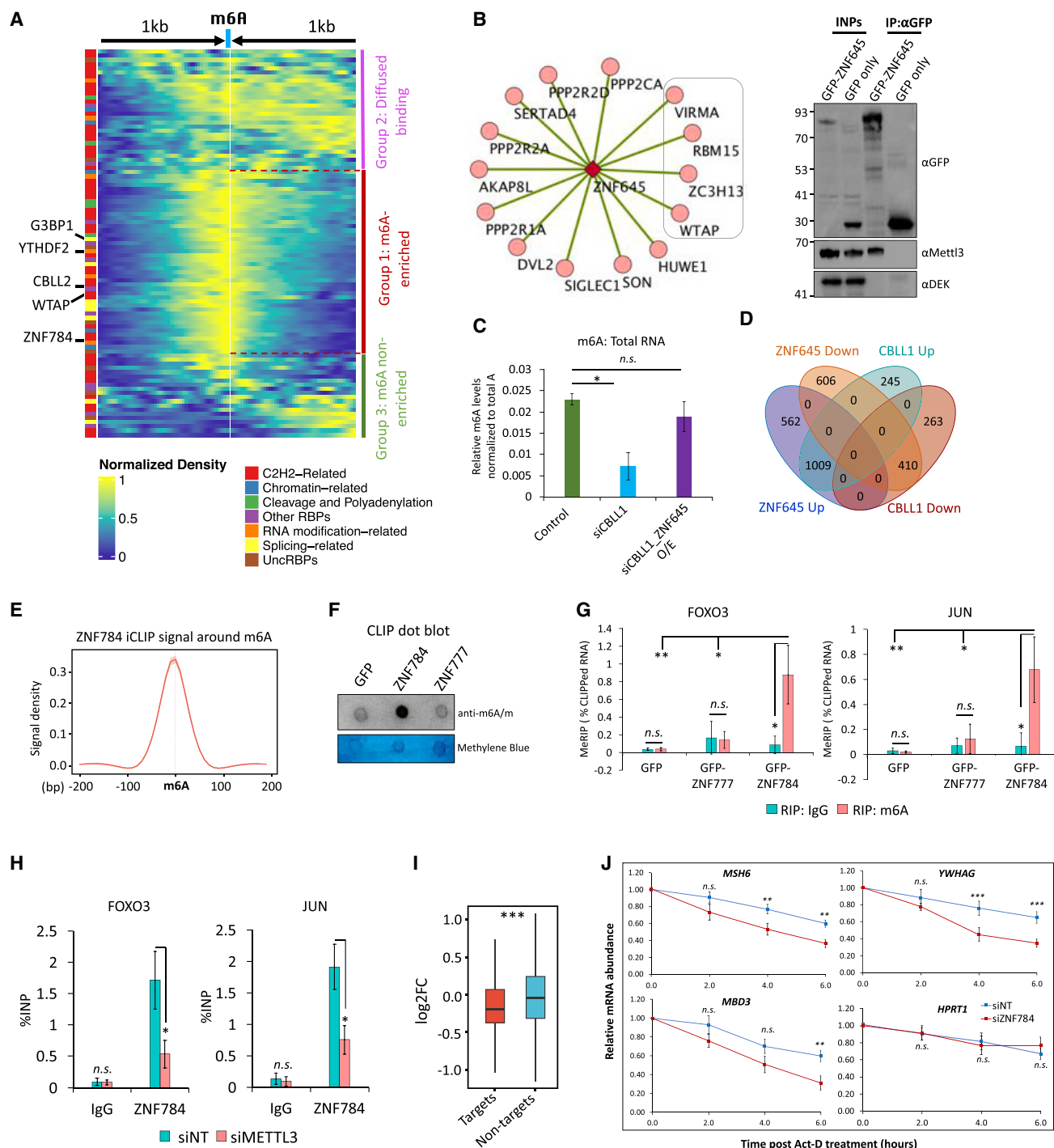


Figure 5. Certain C2H2-ZNFs are enriched around m⁶A sites

(A) Heatmap representation of C2H2-ZNF iCLIP-seq signals around m⁶A sites. Normalized iCLIP-seq peak densities for 1 kb on each side of the m⁶A sites are colored in the same way as in [Figure 2](#). Groups are manually labeled.

(B) Left, protein-protein interaction network of ZNF645 (BFDR ≤ 0.01). m⁶A methyltransferase complex subunits are boxed. Right, coIP experiments using GFP-ZNF645 as a bait. Western blots were probed with the indicated antibodies.

(C) m⁶A quantification using RNA-MS/MS in HEK293 cells treated with the indicated siRNAs (Student's t test, **p* ≤ 0.05).

(D) Venn diagram showing the overlap of differentially expressed genes in either ZNF645- or CBLL1-overexpressing cells.

(E) Metaplot for ZNF784 iCLIP-seq signals indicating the enrichment around target m⁶A sites.

(legend continued on next page)

we examined binding of C2H2-ZNFs around m⁶A sites (Figures 5A and S17) identified by m⁶A-iCLIP-seq (miCLIP-seq) in RNA extracted from HEK293 cells (Figures S16A–S16C; Table S12). For these analyses, we only considered sites with the DRACH (D = A, G, U; R = A, G; H = A, C, U) recognition motif for m⁶A. Clustering revealed three groups of proteins (Figure 5A): group 1, with highly localized enrichment around m⁶A sites; group 2, with weak enrichment in a broader region around m⁶A; and group 3, with no enrichment around m⁶A. Notably, two well-documented direct m⁶A readers, YTHDF1 and YTHDF2, were present in group 1 (Figures 5A and S17), and ~20 C2H2-ZNFs were also in group 1, suggesting that they might also serve as m⁶A regulators (Figure S17). We also used the recently reported single-base-resolution m⁶A sites identified using the glyoxal- and nitrite-mediated deamination of unmethylated adenosines (GLORI) method, which, unlike miCLIP-seq, does not rely on antibodies to detect methylated sites. The same set of C2H2-ZNFs (and other RBPs) exhibited enrichment around GLORI-identified m⁶A sites in HEK293 cells³⁹ (Figure S18), ruling out the possibility that observed enrichments were solely due to methodological similarities of miCLIP- and iCLIP-seq.

These analyses do not necessarily imply, however, that proteins enriched near m⁶A sites bind m⁶A-modified residues directly, and, consistently, the m⁶A-repellent protein G3BP1 also exhibited enrichment around m⁶A sites, as previously reported.⁴⁰ Because observed enrichments might be due to localized binding of proteins within certain mRNA regions, we separately examined enrichment of C2H2-ZNFs around m⁶A sites in 5' UTRs, CDS, introns, and 3' UTRs. Again, the same set of C2H2-ZNFs exhibited enrichment around m⁶A sites in each mRNA feature (Figure S19). Although the signals were centered near m⁶A sites in CDS and ~200 nt upstream of m⁶A sites in 3' UTRs, and they were somewhat sparse in 5' UTRs and introns, these analyses suggest that observed enrichments are not likely solely due to localized enrichment of RBPs. Nevertheless, we cannot rule out the possibility that the observed enrichment for some proteins might be due to their localized binding in the same mRNA regions as m⁶A sites.

ZNF645 (or CBLL2) is a testes-specific paralog of methyltransferase complex (MTC) subunit CBLL1 (Hakai) and clustered with MTC subunit WTAP in group 1 (Figures 5A and S20A–S20C). Our AP-MS analysis identified MTC subunits WTAP, KIAA1429 (VIRMA), ZC3H13, and RBM15, but not the catalytic subunit METTL3, as high-confidence interaction partners of ZNF645 (FDR ≤ 0.01) (Figures 5B and S20D; Table S13). ZNF645 interaction with endogenous METTL3 was, however, identified using colIP (Figure 5B, right), suggesting that ZNF645 might regulate

m⁶A deposition. Importantly, interactions of ZNF645 with METTL3 and WTAP were not observed when colIPs were performed under CLIP buffer conditions using UV-crosslinked cell lysates (Figure S20E). Mass spectrometry experiments quantifying m⁶A in total cellular RNA showed that CBLL1 depletion causes significant reduction in m⁶A that is rescued by ZNF645 overexpression (Figures 5C and S20F), consistent with ZNF645 and CBLL1 being functionally redundant. These findings were validated for selected targets using m⁶A-RNA immunoprecipitation (RIP) experiments (Figure S20G).

We also performed RNA-seq analysis in HEK293 cells overexpressing ZNF645 or CBLL1, finding that ~2,000 genes exhibited differential expression in each condition (log₂ fold change ± 1; adjusted *p* ≤ 0.05) (Tables S14 and S15). Most differential genes in ZNF645-overexpressing cells overlapped with those identified in CBLL1-overexpressing cells (Figure 5D). However, ZNF645-dependent differential genes overlapped very poorly with the ChIP-seq targets of ZNF645 (Figure S20H), suggesting that ZNF645-mediated gene expression alterations are related to the role of ZNF645 in m⁶A regulation rather than its putative TF function(s). Consistently, the m⁶A-containing, ZNF645-bound, transcripts exhibited significantly reduced expression levels compared with ZNF645 non-targets (Figure S20I).

We then examined an uncharacterized group 1 protein, ZNF784 (Figures 5A and 5E), similarly enriched around m⁶A sites. Consistently, CLIP-m⁶A-dot blot assays demonstrated that m⁶A-modified RNAs were co-immunoprecipitated with ZNF784 (Figure 5F), whereas no appreciable m⁶A signal co-precipitated with EGFP alone or ZNF777. UV crosslinking followed by m⁶A-RIP-qPCR confirmed that, in contrast to EGFP and ZNF777, precipitation of crosslinked ZNF784 strongly enriched m⁶A for two targets we examined (Figure 5G). Moreover, ZNF784 CLIP-qPCR indicated that ZNF784 binding to the same two target mRNAs was significantly reduced in siMETTL3 cells (Figures 5H and S21A), indicating that RNA binding by ZNF784 is increased by the presence of m⁶A.

To assess the effect of ZNF784 on expression of m⁶A-containing mRNAs, RNA-seq was performed following knockdown of ZNF784 with siRNA (Figure S21A; Table S16), and six of the expression changes (three increases; three decreases) were validated by RT-qPCR using a different siRNA (siZNF784 # 2) (Figure S21B). Global analysis of expression changes indicated that m⁶A-containing transcripts became slightly less abundant, on average, and therefore almost certainly less stable, in comparison with non-m⁶A genes in siZNF784 samples (Figure 5I). Overlapping ZNF784 CITS with DRACH-motif-containing m⁶A sites then identified ~400 mRNAs bound by ZNF784 within

(F) CLIP RNA was blotted as a dot on a positively charged membrane. The membrane was probed with anti-m⁶A antibody. Methylene blue was used as a stain to indicate the total RNA in each dot.

(G) CLIP was performed for ZNF784, ZNF777, and GFP, and the isolated RNA was subjected to RIP using either anti-m⁶A antibody or IgG. The resulting RNA material was subjected to RT-qPCR analysis using primers designed around known m⁶A peaks for two targets. Enrichment is shown as %Input (CLIP RNA).

(H) ZNF784 CLIP-qPCR analysis using cells treated either with siMETTL3 or siNT.

(I) Boxplot representation of the average expression levels in siZNF784 for m⁶A-containing targets, in comparison with non-m⁶A (non-targets) (post hoc Tukey's honest significant difference (HSD) test, ****p* ≤ 0.001).

(J) RT-qPCR analysis of mRNAs for the indicated genes after treating siNT or siZNF784 cells with actinomycin D for the indicated times. Note: statistical significance in all RT-qPCR experiments was assessed using Student's *t* test (****p* ≤ 0.001, ***p* ≤ 0.01, **p* ≤ 0.05, n.s.: non-significant). Error bars represent standard error of mean (SEM) and *n* = 3.

10 nt of at least one m⁶A site (Figure S21C; Table S17). To determine whether ZNF784 binding near m⁶A modulates mRNA stability, we blocked pre-mRNA synthesis with the transcription inhibitor actinomycin D and followed transcript levels over time. Transcripts from six ZNF784 target genes, MSH6, YWHAG, MBD3, CTDSP2, BRCA1, and MCM5, were significantly destabilized, as expected, in ZNF784-depleted cells, whereas JUN transcripts were stabilized, again as expected, in comparison with control cells (Figures 5J and S21D). No significant difference in stability was observed for mRNA from the non-target gene HPRT1. ZNF784 binding to certain m⁶A-marked target mRNAs is important for regulating mRNA stability for genes specifically enriched in cell-cycle and splicing regulation (Figure S21E). Because our AP-MS data did not reveal interactions of ZNF784 with any known m⁶A-related proteins (Table S18), further studies will be needed to investigate how ZNF784 modulates the stability of certain functionally important m⁶A-containing transcripts.

DISCUSSION

We have systematically identified RNA-binding sites for a large number of C2H2-ZNFs and various other RBPs. Whereas C2H2-ZNFs are traditionally thought to be DNA-binding TFs, we found that C2H2-ZNFs generally also bind RNA. By integrating genomic with proteomic datasets and carrying out biochemical studies, we found that RNA binding by C2H2-ZNFs is usually independent of their DNA binding and that C2H2-ZNFs participate in diverse post-transcriptional regulatory pathways. Given that most RNA-binding sites fell within introns, most C2H2-ZNFs likely bind nascent transcripts in nuclei. In this regard, future efforts should utilize a fractionation iCLIP-seq approach to conclusively establish that C2H2-ZNFs usually bind nascent RNA. Although some non-productive transient interactions with RNA remain possible, our localization of most C2H2-ZNFs to specific mRNA regions suggests that there is substantial sequence/structural specificity in RNA binding by C2H2-ZNFs.

Previous proteomic studies suggested that certain chromatin-associated proteins and TFs bind RNA.^{17–20} Moreover, mRNA-interactome capture studies identified hundreds of putative RBPs that lack classical RNA-binding domains, including TFs.^{17,19} As these earlier investigations exclusively captured RBPs linked to processed mRNAs,^{17,19} only a few C2H2-ZNFs were identified among those expressed in the cell types employed in mRNA-interactome studies. Interestingly, a recent study describing capture of the RNA-bound proteome identified many TFs, including some C2H2-ZNFs, crosslinking to RNA.³² They also identified an RNA-binding motif near the DNA-binding domains of many TFs, somewhat resembling the arginine-rich motif (ARM) of the RNA-binding HIV transcriptional activator Tat.³² We recently showed that a truncation mutant of the C2H2-ZNF SP1 containing its ZNF domains crosslinks robustly to cellular RNA,¹⁴ and examining its sequence revealed stretches of positively charged amino acids (e.g., RRTRR) just upstream of its three ZNF domains (Figure S22).

Although RNA binding by C2H2-ZNFs is mostly distinct from their DNA binding, the observed overlaps, albeit small, could reflect substantial numbers of genomic sites, and these overlapping binding events might be functionally relevant. For example,

RNA binding at certain genomic loci might be important for maintaining proper chromatin architecture through long-range interactions, as was previously shown for CTCF.^{41,42} RNA binding by C2H2-ZNFs might also boost their DNA binding, or vice versa, as when YY1 binding to nascent RNAs acts as a kinetic sink to increase YY1 occupancy on enhancers and promoters.⁹ Our analysis of previously published quantitative proteomic data²⁴ also suggested that there is crosstalk between the transcriptional and post-transcriptional functions of these proteins.

Many studies have revealed RNA binding by specific TFs.^{9,20,24,32,42,43} However, a recent report cast doubt on RNA binding by chromatin-related proteins.⁴⁴ This study primarily challenged the widely reported RNA binding by PRC2, attributing it to experimental artifacts by showing that denaturing purification conditions abolished the interaction of PRC2 with RNA.⁴⁴ Although certain chromatin-related proteins were found to directly bind RNA in cells, YY1 and CTCF, also previously shown to interact with RNA,^{11,13,45,46} did not exhibit RNA signals under denaturing purification conditions.⁴⁴ Although some chromatin-related proteins may not bind RNA *in vivo*, it is important not to generalize these results due to the study's small sample size. Furthermore, these findings should not undermine numerous rigorously performed studies demonstrating the RNA-related functions of certain TFs, including SP1, YY1, and CTCF.^{9,11,14,22,41,42} For example, specific mutations in RNA-binding regions abrogate CTCF-dependent chromatin loops.⁴² Our study indicates that C2H2-ZNFs do exhibit robust RNA signals under both denaturing and regular CLIP purification conditions. We also observed shifts to lower apparent molecular weights in autoradiographs of C2H2-ZNF-associated RNAs when treated with high RNase I concentrations, indicative of direct RNA binding *in vivo*. Finally, RNA binding by CTCF, YY1, and other C2H2-ZNFs in our study depended on their presence in the cells that were UV-irradiated, eliminating the possibility that RNA signals were a consequence of protein-RNA or protein-protein associations occurring after cell lysis.

In a thorough examination of the stringency of the standard iCLIP procedure, which includes highly rigorous washes followed by SDS-PAGE, we found that crosslinking signals depended on UV, and did not occur with control proteins, and that the iCLIP procedure effectively eliminated non-covalent *in vitro* associations with RBPs and/or RNA. In our AP-MS studies performed using CLIP buffer conditions, the presence of co-purifying RBPs was negligible, reinforcing the specificity of our iCLIP-seq pipeline. Our use of a uniform platform also enabled us to eliminate background sites of non-specific crosslinking and signals from RNA binding by antibody chains, and the crosslinking hotspots we identified should prove useful in future CLIP-based studies. Although denaturing purification conditions can also remove proteins and RNAs that interact strongly *in vitro*,⁴⁵ we propose that regular CLIP purification conditions, if implemented as rigorously as recommended,^{16,45} suffice to eliminate non-specific interactions and that denaturing purification conditions should be employed only on a case-by-case basis, such as when dealing with very strong interaction partners.

Although a systematic analysis is currently lacking, C2H2-ZNFs are thought to be predominantly nuclear (Human Protein

Atlas: <https://www.proteinatlas.org/>), consistent with our finding that most crosslink sites are located in introns. Also, consistently, C2H2-ZNFs appear to be involved in many different nuclear post-transcriptional processes, including AS, APA,¹⁴ mRNA export,⁴⁶ and m⁶A regulation. Certain C2H2-ZNFs might even participate in multiple RNA-related processes. Alternatively, multiple C2H2-ZNFs might regulate a single post-transcriptional process by interacting with each other and/or other RBPs. We previously screened genetically for mouse AS regulators and found that mouse C2H2-ZNFs might control AS events linked to cell fate.⁴⁷ The RNA binding around splice sites observed here for many C2H2-ZNFs also suggests that many human C2H2-ZNFs might directly regulate AS.

Our analyses also potentially implicated many C2H2-ZNFs and other RBPs in m⁶A regulation, although some proteins likely showed enrichment around m⁶A sites merely because of their inherent binding preferences across mRNAs. In addition to YTH (YT521-B homology) proteins, several other context-specific m⁶A readers, as well as m⁶A repellent proteins, have been identified.^{40,48} Although our analysis is correlational and does not definitively indicate binding to m⁶A, the observed enrichment of many C2H2-ZNFs around m⁶A sites is intriguing. For example, ZNF784 may be an m⁶A reader protein that enhances the stability of most of its target m⁶A-modified RNAs. Although the precise mechanism remains unclear, we propose that ZNF784 might compete with other nuclear m⁶A readers (e.g., YTHDC1), shielding its targets from m⁶A-mediated destabilization. Two unconventional RBPs, H1FO and LYAR, also exhibited highly localized enrichment around m⁶A sites in our experiments, and linker histone H1 was recently shown to regulate non-coding RNA turnover in an m⁶A-dependent manner.⁴⁹ We speculate that at least several of the C2H2-ZNFs and other proteins that exhibited enrichment around m⁶A sites might function as context-specific m⁶A readers and/or repellents.

Our findings also indicate that ZNF645 (CBLL2), which interacts with the m⁶A writer complex, can functionally substitute for its Hakai subunit. Considering that ZNF645 expression is testis-specific, further studies will be required to explore whether ZNF645 play a role in recruiting the m⁶A machinery to tissue-specific targets. We also recently showed that another C2H2-ZNF, ZBTB48, recruits the m⁶A demethylase FTO to modulate cellular m⁶A levels.⁵⁰ Therefore, particular C2H2-ZNFs can interact with m⁶A writer and eraser proteins.

Our observation that many C2H2-ZNFs target 3' UTRs suggests roles for these proteins in APA regulation and/or mRNA stability. Although we reported that SP1 controls APA for many genes,¹⁴ other C2H2-ZNFs might regulate APA for smaller numbers of specific genes. Although our forced expression of tagged C2H2-ZNFs might cause certain nuclear C2H2-ZNFs to be present in the cytosol, where they might crosslink preferentially to 3' UTRs (which, unlike coding regions, are not protected by ribosomes), our SP1 iCLIP-seq, performed utilizing antibodies against endogenous SP1, excluded that possibility. In addition, we found that various exogenously expressed, GFP-tagged regulators of splicing and CPA (e.g., SRSF1 or CPSF1) localized preferentially to their normal regions on mRNAs and that their binding sites were enriched for motifs reported elsewhere in the literature.

Although further studies are needed to thoroughly explore post-transcriptional regulation by C2H2-ZNFs, we envision that the extensive C2H2-ZNF iCLIP-seq data presented here will provide a unique resource to guide future studies and serve as a framework for RNA researchers to further explore the principles of post-transcriptional regulation.

Limitations of the study

Although our analysis implicates C2H2-ZNFs in multiple post-transcriptional processes, including AS and m⁶A regulation, further studies are needed to both explore the extent to which C2H2-ZNFs are involved in post-transcriptional processes and dissect the molecular mechanisms and biological relevance of RNA binding by these proteins. For example, it will be interesting to explore whether and how ZNF645 regulates m⁶A installation on specific transcripts. Although our studies demonstrate RNA binding is prevalent among C2H2-ZNFs, we did not identify the protein regions (or domains) responsible for their interactions with RNA. Furthermore, we did not explore whether there is any correlation between various accessory domains (and/or number of ZnFs) in C2H2-ZNFs and their RNA-binding preferences. Moreover, our analysis need not imply direct recognition of m⁶A and does not preclude enrichment around m⁶A sites being due to region-specific binding preferences in certain cases. Although our iCLIP-seq experiments used highly stringent conditions, and we adopted several experimental and computational approaches to minimize false positives in our data, some non-productive transient interactions remain a possibility. Although some C2H2-ZNFs were shown to function in post-transcriptional regulation in conjunction with other RBPs, we did not explore the global co-regulatory potential of C2H2-ZNFs and their interacting RBPs. Moreover, it will be important to conduct SRSF1 and ZNF768 knockdown experiments, both individually as well as co-depletion studies, followed by RNA-seq analysis to further explore the co-regulatory potential of these two proteins in AS regulation.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources, reagents, and materials should be directed to and will be fulfilled by the lead contact, Jack F. Greenblatt (jack.greenblatt@utoronto.ca).

Materials availability

All reagents generated in this study are available from the [lead contact](#) with a completed Materials Transfer Agreement.

Data and code availability

- The mass spectrometry data reported in this paper are at (<https://massive.ucsd.edu/>). The accession number is MassIVE: MSV000091917. ChIP-seq, iCLIP-seq, and RNA-seq data generated in this paper are being deposited at Gene Expression Omnibus (GEO, <https://www.ncbi.nlm.nih.gov/geo/>) with the accession code GEO: GSE230846 for series. These data are publicly available as of the date of publication.
- Scripts for the icSHAPE/iCLIP overlap pipeline are publicly available through GitHub: https://github.com/jburnsuoft/icSHAPE_iCLIP. Scripts for GenomicPlot are also publicly available through GitHub: <https://github.com/shuye2009/GenomicPlot>. Other custom analysis scripts are available upon request.

- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

S.N.-S. and J.F.G. conceived and designed the study. S.N.-S. performed all experiments (with help from others when mentioned below), participated in the computational analyses, and contributed to study design. S.P. performed computational analyses related to iCLIP-seq, AP-MS, and RNA-seq data, with input from S.N.-S. J.D.B. performed analyses related to icSHAPE and RNA structural predictions. U.B. and H.L. participated in iCLIP-seq data analysis. N.A. participated in iCLIP and colPs. G.L.B. performed knockdowns and qPCRs. E.R., G.Z., H.T., and E.M. took part in AP-MS. Z.Z. supervised H.L. T.R.H. edited the manuscript. B.J.B. participated in data analysis and edited the manuscript. J.F.G. supervised S.N.-S., S.P., J.D.B., N.A., G.L.B., E.R., G.Z., H.T., and E.M., and supervised the project. S.N.-S. wrote the manuscript with edits and feedback from J.F.G. All authors have read and approved the final manuscript.

DECLARATION OF INTERESTS

B.J.B. is a member of the *Molecular Cell* advisory board.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal Anti-GFP	Abcam	ab290; RRID: AB_303395
Mouse monoclonal Anti-GFP	Life Technologies	G10362; RRID: AB_2536526
Mouse monoclonal Anti-Flag M2	Sigma-Aldrich	F3165; RRID: AB_262044
Anti-SC35	Abcam	ab204916; RRID: AB_2909393
Anti-m ⁶ A	MBL	RN131P
Anti-m ⁶ A	Synaptic Systems	202003; RRID: AB_2279214
Mouse monoclonal Anti-GAPDH	Abcam	ab8245; RRID: AB_2107448
Normal mouse IgG	SANTA Cruz BioTech.	sc-2025; RRID: AB_737182
Anti-SF3B1	Thermo Fisher Scientific	PA5-41723; RRID: AB_2610040
Anti-U1-70K	Abcam	ab83306; RRID: AB_10673827
Anti-METTL3	Abcam	ab195352; RRID: AB_2721254
Anti-DEK	Thermo Fisher Scientific	16448-1-AP; RRID: AB_2092097
Anti-U2AF1	Thermo Fisher Scientific	PA5-28510; RRID: AB_2545986
Anti-WT1 (H-1)	Santa Cruz Biotechnology	sc-393498; RRID: AB_2905496
Anti-TRIM28	Invitrogen	MA1-2023; RRID: AB_2209892
Rabbit polyclonal Anti-YTHDF2	Proteintech	Cat# 24744-1-AP; RRID: AB_2687435
Goat anti-Rabbit IgG (H+L) Secondary Antibody, HRP	Thermo Fisher Scientific	Catalog # 31460; RRID: AB_228341
Goat anti-rabbit IgG Secondary Antibody, HRP	DAKO	P0448; RRID: AB_2617138
Goat anti-Mouse IgG (H+L) Secondary Antibody, HRP	Thermo Fisher Scientific	Catalog # 31430; RRID: AB_228307
Bacterial and virus strains		
Subcloning Efficiency DH5a Competent Cells	Thermo Fisher Scientific	18265017
Chemicals, peptides, and recombinant proteins		
4% Paraformaldehyde	VWR	102091-906
Benzonase Nuclease	Sigma-Aldrich	E826
BSA	NEB	B9000S
DAPI	Sigma-Aldrich	D9542-5MG
DMEM (high glucose)	Sigma-Aldrich	D5796
Doxycycline hyclate	Sigma-Aldrich	D9891
FastAP	Thermo Fisher Scientific	EF0651
FBS	GIBCO	12483-020
Hygromycin B	Sigma-Aldrich	10843555001
IGEPAL CA-630	Sigma-Aldrich	I8896-100ML
Lipofectamine 2000	Thermo Fisher Scientific	11668019
Opti-MEM I Reduced Serum Medium	GIBCO	31985070
Proteinase K	Thermo Fisher Scientific	25530049
Poly-D-Lysine	Sigma-Aldrich	6407
Puromycin	Sigma-Aldrich	P8833
RNAiMAX	Thermo Fisher Scientific	13778150
Lipofectamine iMax transfection reagent	Thermo Fisher Scientific	13778075
RNase I	Thermo Fisher Scientific	AM2295
RNaseOUT Recombinant Ribonuclease Inhibitor	Invitrogen	10777019
Sodium pyruvate	GIBCO	11360-070
SUPERase In RNase Inhibitor	Ambion	AM2696

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Triton X-100	Sigma-Aldrich	T8787-100ML
Turbo DNase	Thermo Fisher Scientific	AM2238
T4 polynucleotide kinase	NEB	M0201L
FastAP Thermosensitive Alkaline Phosphatase	Thermo Fisher Scientific	EF0651
Sodium arsenite	Sigma-Aldrich	N/A
Actinomycin D	Sigma-Aldrich	A1410
Trizol Reagent	Invitrogen	15596018
β-mercaptoethanol	Sigma-Aldrich	M6250
N6-Methyladenosine 5'-monophosphate sodium salt	Sigma-Aldrich	M2780
Glycogen	Sigma-Aldrich	CAT #G1767
NuPAGE sample reducing agent	Life Technologies	NP0004

Critical commercial assays

Accuprime SuperMix I	Thermo Fisher Scientific	12344040
AllPrep DNA/RNA Mini Kit	QIAGEN	80204
Dynabeads protein G	Thermo Fisher Scientific	10004D
Maxima First Strand cDNA Synthesis Kit for RT-qPCR	Thermo Fisher Scientific	K1671
OneStep RT-PCR Kit	QIAGEN	210212
NEBNext High-Fidelity 2X PCR Master Mix	NEB	M0541L
SuperScript VIL0 Kit	Invitrogen	11754
SYBR Green Master qPCR mix	Applied Bioscience	4385617
RNeasy Mini Kit	QIAGEN	74106
Select-a-Size DNA Clean & Concentrator	Zymo	D4080
CircLigase™ II ssDNA Ligase	Lucigen	CL9025K
EpiQuik m ⁶ A RNA Methylation Quantification Kit	Epigentek	CAT #P-9008-48
Dynabeads mRNA Purification Kit	Thermo Fisher Scientific	CAT #61006
SuperSignal™ West Pico PLUS Chemiluminescent Substrate	Thermo Fisher Scientific	CAT #34580
Trypan Blue Stain (0.4%) for use with the Countess™ Automated Cell Counter	Invitrogen	Cat #T10282
Enhanced chemiluminescence	GE Healthcare	RPN2232
RNA Fragmentation Reagent	Invitrogen	AM8740
Gibson Assembly Master Mix	NEB	M5510
Pierce™ Silver Stain Kit	Thermo Fisher Scientific	24612
Dynabeads Antibody Coupling Kit	Life Technologies	14311D

Deposited data

iCLIP-seq	This study	GEO: GSE228608
AP-MS	This study	MassIVE: MSV000091917
RNA-seq	This study	GEO: GSE228608

Experimental models: Cell lines

Human: Flp-In 293 T-REx cell lines	Invitrogen	R780-07
HEK293 cells	ATCC	CRL1573

Oligonucleotides

Primers	This study	Table S19
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Recombinant DNA

CRISPR/Cas9 plasmid LentiCRISPRv2	This study	Dr. Jason Moffat; Addgene Cat#52961
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Software and algorithms

MetaPlotR	Olarerin-George and Jaffrey ⁵¹	https://github.com/olarerin/metaPlotR
ShinyGO (v0.66)	Ge et al. ⁵²	http://bioinformatics.sdstate.edu/go/

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
ClusterProfiler 3.16.1	Yu et al. ⁵³	https://bioconductor.org/packages/release/bioc/html/clusterProfiler.html
EnhancedVolcano	(Unpublished)	https://github.com/kevinblighe/EnhancedVolcano
DESeq2 3.11	Love et al. ⁵⁴	https://www.bioconductor.org/packages/release/bioc/html/DESeq2.html
Cytoscape	Cline et al. ⁵⁵	Version: 3.4.0
ImageJ	NIH	https://imagej.nih.gov/ij/
SAINTexpress	Teo et al. ⁵⁶	Version exp3.3
GraphPad Prism	GraphPad	https://www.graphpad.com
GenomicPlot	(Shuye et al., Unpublished)	https://doi.org/10.5281/zenodo.13308113
Tophat 2.1.0	Trapnell et al. ⁵⁷	https://ccb.jhu.edu/software/tophat/
STAR	Dobin et al. ⁵⁸	https://github.com/alexdobin/STAR
RSEM	Li and Dewey ⁵⁹	https://deweylab.github.io/RSEM/
Trimmomatic	Bolger et al. ⁶⁰	http://www.usadellab.org/cms/?page=trimmomatic
CLIP Tool Kit (CTK)	Shah et al. ⁶¹	http://zhanglab.c2b2.columbia.edu/index.php/CTK
MACS2 (version 2.1.2)	Feng et al. ⁶²	https://github.com/taoliu/MACS
BEAM tool (V.1.6.1)	Pietrosanto et al. ⁶³	http://beam.uniroma2.it/download
vast-tools 1.0	Tapial et al. ⁶⁴	https://github.com/vastgroup/vast-tools
icSHAPE/iCLIP overlap	This study	https://doi.org/10.5281/zenodo.13375429

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell culture

HEK293 Flp-In T-Rex cell lines were purchased from Life Technologies (Invitrogen catalogue number R780-07). Cell cultures were maintained in Dulbecco's modified Eagle's medium (DMEM) (Wisent Bioproducts catalogue number 319-005-CL) supplemented with 10% FBS (Wisent Bioproducts catalogue number 080-705), sodium pyruvate, non-essential amino acids, and penicillin/streptomycin as described.⁶⁵

METHOD DETAILS

Epitope tagging in HEK293 cells

Gateway™-compatible entry clones were cloned into the pDEST pcDNA5/FRT/TO-eGFP vector according to the manufacturer's instructions. The vector was co-transfected into HEK293 Flp-In T-Rex cells with the pOG44 Flp recombinase expression plasmid. Cells were selected with hygromycin (Life Technologies, 10687010) at 200 µg/ml for FRT site-specific recombination into the genome. All cell lines were routinely monitored for mycoplasma contamination.

Expression was induced by adding Doxycycline (1 µg/ml) to the culture medium 24 hours prior to harvesting the cells. HEK293 Flp-In T-Rex cell lines for the GFP-tagged C2H2-ZNF proteins were described previously in our reports.^{8,21}

Endogenous C-terminal EGFP-tagging

DEK, SRSF1, SSBP1, CTCF, RPRD1A, and PRDX1 were endogenously EGFP-tagged at their C-termini in HEK293 cells using CRISPR-directed targeted genomic insertion. HEK293 cells were co-transfected with pX330 (or pX334) plasmid expressing hSpCas9 and an sgRNA targeting the gene of interest near the stop codon, as well as with a repair template, using Lipofectamine 2000 (Invitrogen). The repair templates containing the EGFP sequence, flanked by 0.5Kb homology arms corresponding to the up- and downstream sequences of the stop codons for each gene of interest, were prepared using Gibson Assembly Master Mix (NEB #M5510) according to the manufacturer's instructions (Figure S23). CRISPR guide RNAs were designed using the Zhang lab tool (available through Addgene: <https://www.addgene.org/crispr/zhang/>) (guide sequences are available upon request).

Cells were selected for 3 days with puromycin (2.5 $\mu\text{g}/\text{mL}$) and cultured for seven more days before single cell sorting for EGFP+ cells with a BD Influx FACS sorter (BD Biosciences). The insertion sequence in each sorted clone was confirmed at the genomic level using Sanger sequencing, and clones were tested for EGFP-tagged proteins' expression using Western blotting.

Immunoprecipitation (IP) and western blots

1 mL of RIPA lysis buffer containing 140 mM NaCl, 10 mM Tris pH 7.6–8.0, 1% Triton X-100, 0.1% sodium deoxycholate, and 1 mM EDTA supplemented with protease inhibitors (Roche catalogue number 05892791001) was used for cell lysis. 75 units of Benzonase (Sigma E1014) was added and cell extracts were incubated for 30 minutes in the cold room with end-to-end rotation. The cell lysates were cleared in a microcentrifuge at 15,000 g for 30 minutes at 4°C. The clarified cell lysate was incubated with 1 μg of GFP antibody (Life Technologies G10362) for 2 hours. After adding 10 μL Protein G Dynabeads (Invitrogen; catalogue number 10003D), lysates were incubated for another 2 hours at 4°C. The samples were washed three times with lysis buffer containing an additional 1% NP40 and 2% Triton X-100 for 5 minutes each at 4°C with end-to-end rotation. The samples were boiled in SDS gel sample buffer, resolved using 4–12% BisTris–PAGE and transferred to a PVDF membrane (Bio-Rad catalogue number 162-0177) using a Gel Transfer Cell (BioRad catalogue number 1703930). Primary antibodies were used at the following dilutions: 1:5000 for GFP (Abcam polyclonal antibody catalogue number 290) and FLAG (Sigma monoclonal antibody catalogue number F1804), and 1:1000 for SF3B1 (Thermo Fisher Scientific PA5-41723), U1-70K (ab83306) and METTL3 (ab195352). Horseradish peroxidase-conjugated goat anti-mouse (Thermo Fisher 31430) or anti-rabbit secondary (Thermo Fisher 31460) antibodies were used at 1:5000. Blots were developed using Pierce ECL Western Blotting Substrate (Thermo Scientific catalogue number 32106).

For silver staining experiments, cell lysis and IPs from UV-crosslinked samples were performed using either RIPA buffer or iCLIP buffer (described below) conditions. Samples were resolved using 10% Tris-Glycine PAGE and gels were stained using a 'Pierce™ Silver Stain Kit' as per the manufacturer's instructions (Thermo Scientific REF 24612).

CLIP-autoradiography and RNase I titration

CLIP-autoradiography was performed essentially the same way as for the initial steps of iCLIP.⁴⁵ HEK293 cells were grown in 15 cm culture plates, and expression of GFP-tagged proteins was induced with Doxycycline (1 $\mu\text{g}/\text{mL}$) for 24 hours. Cells were UV cross-linked with 0.15 J/cm² at 254 nm in a Stratalinker 1800. Following lysis in 2.2 mL of iCLIP lysis buffer, 1 mL cell lysate was incubated with 4 μL Turbo DNase and 20 μL of varying dilutions of RNase I (e.g., 1:5, 1:50, 1:250, 1:500, 1:1000, 1:5000) for 5 minutes at 37°C while shaking in a thermomixer. GFP-tagged proteins were immunoprecipitated using 6 μg of anti-GFP antibody (Life Technologies G10362) and 60 μL Protein G Dynabeads (Invitrogen). Beads were washed 6 times with iCLIP high salt buffer (5-minute end-to-end rotation in a cold room for each wash) followed by three washes with PNK buffer. The immunoprecipitated RNA was 5'-end-labeled with ³²P using T4 polynucleotide kinase (New England Biolabs catalogue number M0201L), separated using 4–12% BisTris–PAGE and then transferred overnight to a nitrocellulose membrane (Protran) in a cold room followed by autoradiography. For initial CLIP-autoradiography experiments (shown in Figure S1A), we used 1:250 RNase I dilution (unless indicated). To perform iCLIP-seq procedure (see below), we carried out RNase I titration to identify RNase I concentration that yields optimal RNA fragmentation (RNA size range should be between 50 and 300 nucleotides). Note: At optimal RNase I concentration, the fragmented RNA bound to the protein is expected to migrate above the protein during SDS-PAGE.

Stringent urea denaturing CLIP

Urea denaturing CLIP was performed using FLAG-tagged proteins, essentially as described previously.⁴⁵ In this procedure, a dual immunoprecipitation with urea denaturation conditions is performed. Briefly, 30 μg of M2 FLAG antibody was covalently coupled to 1 mg of Dynabeads M-270 Epoxy according to the manufacturer's instructions (Dynabeads Antibody Coupling Kit, Life Technologies). Beads were resuspended in 100 μL SB (contained in Dynabeads Antibody Coupling Kit) per 1 mg beads. 40 μL of the beads were utilized for each purification. Cell lysis, Turbo DNase treatment, partial RNase I digestion, and lysate clarification steps were carried out as described above. Cell lysates were incubated with M2 beads for 4 hours in a cold room. Beads were washed 6 times with iCLIP high salt buffer (5-minute end-to-end rotation in a cold room for each wash) followed by three washes with PNK buffer. 100 μL urea cracking buffer was then added to the beads, and the beads were incubated at 65 °C, with shaking at 1100 rpm, for 3 min to elute the immunoprecipitated proteins. 900 μL T-20 IP buffer supplemented with RNase and Protease inhibitors was added. This material was subsequently incubated with 5 μg of FLAG antibody (Sigma) coupled with Protein G Dynabeads (Invitrogen) for 2–4 hours with end-to-end rotation at 4°C. Beads were washed 3 times with iCLIP high salt buffer (5-minute end-to-end rotation in a cold room) followed by three washes with PNK buffer. Subsequent steps, including radiolabelling, elution, SDS-PAGE, and autoradiography were performed essentially as described above.

iCLIP-seq procedure

Individual nucleotide resolution UV crosslinking and immunoprecipitation (iCLIP) was performed with the modifications implemented in our previous reports.^{14,65} Briefly, HEK293 cells were grown in 15 cm culture plates in two independent batches representing biological replicates. Expression of GFP-tagged proteins was induced with Doxycycline (1 $\mu\text{g}/\text{mL}$) for 24 hours and UV cross-linked with 0.15 J/cm² at 254 nm in a Stratalinker 1800. Following cell lysis in 2.2 mL of iCLIP lysis buffer, 1 mL of the lysate was incubated with Turbo DNase (Life Technologies catalogue number AM2238) and RNase I (Ambion catalogue number AM2294) for 5 minutes at 37°C

to digest genomic DNA and obtain RNA fragments of an optimal size range (between 50 and 300 nucleotides). Optimal RNase I dilution for each protein was experimentally determined prior to performing iCLIP-seq. For each experiment, cell lysates were pre-clari- fied by incubating with 50 μ L of Protein G Dynabeads (Invitrogen) for 2 hours in a cold room prior to IPs. GFP-tagged proteins were immunoprecipitated using 6 μ g of anti-GFP antibody (Life Technologies G10362) and 60 μ L Protein G Dynabeads (Invitrogen) for 2–4 hours (to overnight). A total of 2% input material (from one iCLIP replicate for a given target) was set aside for size-matched control libraries (SMI) prior to the IPs. Beads were washed 5 times with iCLIP high salt buffer (5-minute end-to-end rotation in a cold room for each wash) followed by three washes with PNK buffer. Dephosphorylation with FastAP was performed for 15 minutes at 37°C while shaking, followed by T4 polynucleotide kinase treatment for an additional 20 minutes, as described in eCLIP procedure.⁶⁶ Beads were washed 3 times with iCLIP high salt buffer (5-minute end-to-end rotation for each wash) followed by three washes with PNK buffer. Pre-adenylated iCLIP adaptors were ligated to the 3'-ends of RNAs using the eCLIP ligation method.⁶⁶ Beads were then washed 3 times with iCLIP high salt buffer (5-minute rotation in a cold room for each wash) followed by three washes with PNK buffer. The immunoprecipitated RNA was 5'-end-labeled with ³²P using T4 polynucleotide kinase (New England Biolabs catalogue number M0201L), separated using 4–12% BisTris–PAGE and then transferred overnight to a nitrocellulose membrane (Protran) in a cold room. For the input sample, which was resolved on 4–12% BisTris–PAGE and transferred to a nitrocellulose membrane alongside the immunoprecipitated RNA, the membrane was cut to match the size of the IP material. RNA was recovered from the membrane using proteinase K (Thermo Fisher catalogue number 25530049) digestion. RNA was reverse transcribed into cDNA using barcoded primers. The cDNA was size-selected (low: 70 to 85 nt, middle: 85 to 110 nt, and high: 110 to 180 nt) using gel extraction (6% TBE-urea gels) and circularized with CircLigase™ II ssDNA ligase to add the adaptor to the 5'-end. Betaine was added in the CircLigase reaction at a final concentration of 1 M, and the mixture was incubated for 2 hours at 60°C. Circularized cDNA was digested at the internal BamHI site for linearization, then PCR amplified using AccuPrime SuperMix I (Thermo Fisher catalog number 12344040) or Phusion High-Fidelity PCR Master Mix (NEB, M0531S). The final PCR libraries were agarose gel purified on purification columns (QIAGEN), and the eluted DNA was mixed at a ratio of 1:5:5 from the low, middle, and high fractions and submitted for sequencing on either an Illumina HiSeq 2500 or Illumina NextSeq 500 platform using a full High-Output v2.5 flow cell to generate single-end 51 nucleotide reads with 40M read depth per sample.

The barcoded primers used for iCLIP-seq are listed in [Table S2](#).

miCLIP-seq procedure

miCLIP-seq experiments were performed essentially as previously described with minor modifications.⁶⁷ Total RNA was extracted from HEK293 cells by using TRIzol and following the manufacturer's protocol. RNA was treated with DNase I, and 12 μ g of total RNA was used for miCLIP. 2 μ L of RNA Fragmentation Reagents (Invitrogen AM8740) was added to RNA suspended in 20 μ L of sterile water and incubated in a heat block at 75°C for exactly 12 minutes. The reaction was stopped by adding 2.2 μ L of stop solution (Invitrogen AM8740). The fragmented RNA was incubated with 10 μ g of m⁶A antibody for 2 hours at 4°C. RNA-antibody complexes were UV crosslinked twice with 0.15 J/cm² at 254 nm in a Stratalinker 1800. 100 μ L of Protein G magnetic beads were added, and samples were rotated in a cold room for 2 hours. Following stringent washes with iCLIP high salt buffer and dephosphorylation with FastAP and T4 polynucleotide kinase, pre-adenylated adaptors were ligated to the 3'-ends of RNAs using the eCLIP ligation method⁶⁶ as detailed above. The immunoprecipitated RNA was 5'-end-labeled with ³²P using T4 polynucleotide kinase (New England Biolabs catalogue number M0201L), separated using 4–12% BisTris–PAGE and transferred to a nitrocellulose membrane (Protran). RNA isolation, cDNA preparation, cDNA size selection, circ-ligation, and library amplification steps were performed essentially as described above for the iCLIP-seq procedure. The final PCR libraries were agarose gel purified on purification columns (QIAGEN), and the eluted DNA was mixed at a ratio of 1:5:5 from the low, middle, and high fractions and submitted for sequencing on an Illumina NextSeq 500 platform using a full High-Output v2.5 flow cell to generate single-end 51 nucleotide reads with 40M read depth per sample.

The barcoded primers used for miCLIP-seq are listed in [Table S2](#).

siRNA treatment, RNA isolation, and qRT-PCR

siRNA knockdown experiments were performed in biological triplicates along with control knockdowns. HEK293 cells were trans- fected using 40–50 nM Silencer Select siRNAs with Lipofectamine iMax transfection reagent (Thermo Fisher 13778075) for 72 hours. Silencer Select control siRNA (Thermo Fisher AM4611) and siRNAs against human genes were purchased from Thermo Scientific (catalogue number 4392420) with the following IDs: for ZNF784, s45060 and s45059 (indicated as siZNF784 # 2); for ZBTB7A, s27989 and s27990 (indicated as siZBTB7A # 2), and for CBLL1, s36538. Moreover, 'ON-TARGETplus' human SRSF1 siRNA were purchased from Dharmacon™ (SMARTpool; Catalog: L-018672-01-0005).

Total RNA was extracted using Trizol Reagent (Invitrogen catalogue number 15596018) according to the manufacturer's instruc- tions. For qRT-PCR, RNA was converted into cDNA using a SuperScript VILO Kit (Invitrogen catalogue number 11754). qPCR was performed with Fast SYBR Green Master qPCR mix (Applied Bioscience 4385617) on an Applied Biosystems 7300 real time PCR System (Thermo Fisher catalogue number 4406984). The qPCR program involved 40 cycles of 95 °C for 15 s and 60 °C for 30 s, and a final cycle (95 °C for 15 s and then 60 °C). Primers are listed in [Table S19](#).

RT-PCR assays

Total RNA was extracted using the RNeasy Mini Kit (QIAGEN). RT-PCR assays were carried out using the OneStep RT-PCR kit (QIAGEN), according to the manufacturer's instructions. In these experiments, 20ng of total RNA was used per 10 μ l of reaction. The number of amplification cycles was 27–32 for all transcripts analyzed. Reaction products were separated on 2%–4% agarose gels. Primers used for RT-PCR reactions are available upon request.

RNA sequencing

Total RNA was extracted from HEK293 cells that had been treated with siRNAs for 72 hours, as described above. For overexpression RNA-seq, expression of proteins was induced for 24 hours using Doxycycline (1 μ g/mL). 1 μ g purified total RNA in two biological replicates was submitted to the Donnelly Sequencing Centre for further processing. Briefly, Poly-A selected, rRNA-depleted libraries were generated using the TruSeq Illumina library preparation kit (Epicenter) according to the manufacturer's recommendations. The quantified pool was hybridized at a final concentration of 2.21 pM and sequenced using the Illumina NovaSeq platform for 300 cycles (paired-end sequencing) to a target depth of 50 M reads.

RNA immunoprecipitation (RIP)

We performed RIP experiments as described previously.⁶⁵ Briefly, cells grown in 10-cm plates were harvested and lysed in 1 mL lysis buffer containing 25 mM Tris HCl pH 7.5, 150 mM NaCl, 0.5% (v/v) NP-40, 1 mM AEBSF and 1 mM DTT, supplemented with RNase and protease/phosphatase inhibitors, and incubated on ice for 30 minutes. Lysates were clarified by centrifugation for 30 minutes at 15,000 g and 4°C. 5% of the material was set aside as input. RNA-protein complexes were immunoprecipitated using 5 μ g of anti-GFP antibody (Life Technologies G10362), conjugated with Protein G Dynabeads (Invitrogen) for 2–3 hours at 4°C with end-to-end rotation. 5 μ g of IgG (Santa Cruz Biotechnology, catalog number: sc-2025) was used as a control. The beads were washed three times with wash buffer containing 25 mM Tris HCl pH 7.5, 150 mM NaCl, 0.05% (v/v) NP-40, 1 mM AEBSF and 1 mM DTT, with 5 minutes of rotation at 4°C for each wash. RNA from the input and IP samples was extracted using Trizol Reagent (Invitrogen catalogue number 15596018) according to the manufacturer's instructions. cDNA was synthesized using the First-strand cDNA synthesis kit (Thermo Fisher Scientific) with random hexameric primers.

CLIP-qRT-PCR

CLIP was performed essentially in the same way as the initial steps of the iCLIP procedure detailed above. Lysates generated from the UV crosslinked cells were treated with Turbo DNase and RNase I (1:250 dilution) for 5 minutes at 37°C. RNA-protein complexes were immunoprecipitated using 5 μ g anti-GFP antibody (Life Technologies G10362) and 50 μ l Protein G Dynabeads. For the control, 5 μ g of IgG (Santa Cruz Biotechnology, catalog number: sc-2025) was used. Following high salt washes (5 $\times$ with end-to-end rotation for 5 minutes in a cold room), proteins were digested with proteinase K, then the immunoprecipitated RNA was extracted using Trizol and subjected to qRT-PCR, as described previously.¹⁴ Input RNA (5%) was also reverse transcribed into cDNA and used to calculate the percent enrichment in the immunoprecipitated samples.

m⁶A dot blot assay

Total RNA was isolated using Trizol (Invitrogen, 15596-018) according to the manufacturer's instructions. mRNA purification was performed in two rounds using the Dynabeads® mRNA DIRECT™ kit (ThermoFisher Scientific, #61006). The m⁶A-dot-blot was performed following a published procedure.⁶⁸ Briefly, the primary rabbit anti-m⁶A antibody (Synaptic Systems, 202003) was applied to the charged nylon-based membrane containing RNA samples. HRP-conjugated Goat anti-rabbit IgG (DakoCytomation, p0448) was added to the blots. The blot was developed with enhanced chemiluminescence (GE Healthcare, RPN2232). The signal density of the dot-blot experiment was quantified by ImageJ software (Media Cybernetics). For the CLIP-m⁶A-dot blot, RNA isolated using the initial steps of the iCLIP procedure was subjected to the m⁶A dot blot assay.

CLIP-MeRIP

CLIP-m⁶A-RIP (CLIP-MeRIP) was performed as described previously,⁶⁹ with minor modifications. Initially, CLIP was performed starting with 300 million HEK293 cells expressing GFP-ZNF784. Cells expressing GFP-alone were used as a negative control. Following IPs, proteins were digested on-beads with proteinase K, and RNA fragments were recovered using Trizol. The purified RNA pellet was dissolved in 20 μ l of RNase-free water, of which 5 μ l was set aside as input. The remaining 15 μ l of the purified RNA was then subjected to MeRIP, which was performed as described below. Briefly, the RNA was incubated for 2 hours at 4°C with anti-m⁶A antibody (Synaptic Systems) in IP buffer (150 mM NaCl, 0.1% (v/v) Nonidet P-40, and 10 mM Tris-HCl, pH 7.4) supplemented with murine RNase inhibitor (NEB; M0314L). The antibody-RNA complex was isolated by incubation with protein A beads (Invitrogen) at 4°C for 2 h. The beads were washed three times in the 1 $\times$ IP buffer and then eluted competitively with an m⁶A monophosphate solution (1 $\times$ IP buffer and 6.7 mM m⁶A, Sigma-Aldrich). RNA in the eluate was isolated using RNA Clean and Concentrator (Zymo Research). The m⁶A immunoprecipitated RNA and input RNA (5 μ l of CLIP elute) were processed as in qRT-PCR.

m6A-IP-qRT-PCR (MeRIP-qPCR)

Isolated mRNA was fragmented into ~300 nt fragments using RNA Fragmentation Reagents, as detailed above in the miCLIP procedure. Five percent of the fragmented mRNAs were kept as input. MeRIP was performed using anti-m⁶A antibody (Synaptic Systems, 202003), essentially as detailed above. IgG (Santa Cruz Biotechnology, catalog number: sc-2025) was used as a control. The m⁶A enrichment was measured using qRT-PCR.

Nucleoside mass-spectrometry analysis

Nucleoside mass-spectrometry (MS) analysis was performed essentially as described in published protocols.^{50,70,71} Briefly, RNA was treated with DNase, then purified using an RNA clean up kit (Zymo kit, R1013, Cedarlane), and resuspended in 20 μ l H₂O. 500 to 1000 ng RNA was used for downstream processing. RNA was de-capped with 0.5 U Cap-Clip enzyme (Cell Script, C-CC15011H) for one hour at 37°C, and samples were further digested with 2 U of Nuclease P1 (N8630, Sigma) overnight at 42 °C in NH₄OAc buffer (10 mM, pH 5.3). 1 U of Alkaline phosphatase (Sigma, P4252) and 0.05M MES was added and the reaction was incubated for another 6 hours at 55 °C. The samples were filtered through 10,000 MW cut off spin filters (AMicon, UFC501024) for 10 min at 16,000 g at 4°C. The final filtrate was injected into the MS.

We used a reverse phase micro-capillary liquid trap column and analytical column for molecular separation using the EASY-nLC 1200 system. A 25mm x 75 μ m silica micro-capillary trap column was packed with 2.5 μ m Synergy Hydro RP (Phenomenex), and a tip for a 100mm x 75 μ m silica micro-capillary analytical column was created with a column puller (Sutter Instruments) and packed with 2.5 μ m Synergy Hydro RP stationary phase (Phenomenex). We used the EASY-nLC system to drive a 30 min organic gradient using buffers A (0.1% formic acid) and B (80% acetonitrile with 0.1% formic acid). The molecules were separated at a flow rate of 250 nL/min (gradient buffer B: 0% to 100% - 2 min, 100% - 23 minutes, 100% to 0% - 1 min, 0% - 2 min). Eluates were sprayed into a Q Exactive HF mass spectrometer (ThermoFisher Scientific). The full scan was performed in 60000 resolution with 3E6 AGC.

The nucleosides were identified and quantified using the nucleoside-to-base ion mass transitions of 282.1 to 150.1 (m⁶A), 268 to 136 (A), 296 to 150 (m⁶Am), and 282 to 136 (Am). The m⁶A was quantified as the ratio of m⁶A to A.

Chromatin immunoprecipitation (ChIP) procedure

Chromatin immunoprecipitation was performed essentially as described in detail in our previous publication.⁵⁰ ~20 million HEK293 cells were grown in independent batches in 15cm plates and were crosslinked in 1% formaldehyde in crosslinking buffer (50 mM HEPES, 100 mM NaCl, 1 mM EDTA, 0.5 mM EGTA, 1% formaldehyde). Expression of EGFP-tagged proteins was induced for 24 hours prior to harvesting. 250 μ l of blocking solution and 2 μ g of GFP antibody (Abcam ab290) were added, and beads were rotated at 4°C for at least 4 hrs to bind antibody to beads. Cell pellets were lysed in 3 mL LB1 buffer (50 mM Hepes, 140 mM NaCl, 1 mM EDTA, 10% glycerol, 0.5% NP-40, 0.25% TX-100), supplemented with protease inhibitors (Roche catalogue number 05892791001). DNA was fragmented using sonication in a BioRuptor (BioRuptor settings: power = high, "on" interval = 30 sec, "off" interval = 30 sec (10 cycles of "on"/ "off"). Dynabeads coupled to antibody were added to the sonicated lysate. Samples were rotated at 4°C overnight. Beads were then washed at least 5X in 1 mL cold RIPA buffer (50 mM Hepes, 500 mM LiCl, 1 mM EDTA, 1% NP-40, 0.7% Na-Deoxycholate). Elution was carried out in 200 μ l elution buffer (50 mM Tris, 10 mM EDTA, 1% SDS) for 15 min at 65°C. Both input (10%) and eluted chromatin samples were incubated overnight (12–16 hrs) in a 65°C water bath to reverse crosslinks. RNA was digested for 2 hours at 37°C with RNase A (0.2 mg/mL final concentration) and samples were then incubated with proteinase K (0.2 mg/mL final concentration) at 55°C for 2 hours. DNA was purified with a DNA purification kit (QIAGEN), and concentration was measured by Qubit. The purified DNA was utilized in qPCR assays. Primers are available upon request.

Affinity purification for mass spectrometry

The AP-MS procedure for HEK293 cells was performed essentially as described.^{8,72} Briefly, ~20 million cells were grown independently in two batches representing two biological replicates. To induce protein expression, 1 μ g/mL doxycycline was added 24 hours prior to harvesting. Cells were lysed in 1 mL of high-salt buffer (10 mM Tris-HCl pH 8.0, 420 mM NaCl, 0.1% NP-40, plus protease/phosphatase inhibitors) with three freeze-thaw cycles. Sonication was performed as detailed in our previous reports.⁸ Subsequently, cell lysates were treated with Benzonase for 30 minutes in a cold room with end-to-end rotation. Cell extracts were centrifuged to pellet and remove any cellular debris. GFP-tagged proteins were immunoprecipitated with 1 μ g of GFP antibody (G10362, Life Technologies) overnight in the cold room followed by a 2-hour incubation with Protein G Dynabeads (Invitrogen). Next day, following 3 washes with high-salt buffer (10mM TRIS-HCl, pH7.9, 300mM NaCl, 0.1% NP-40) with end-to-end rotation in the cold room and 2 washes with buffer without detergent (10mM TRIS-HCl, pH7.9, 100 mM NaCl), the immunoprecipitated proteins were eluted with 0.5M NH₄OH and lyophilized.

Affinity purification (UV-crosslinking) followed by mass spectrometry

For EGFP-tagged protein expression, cells were incubated for 24 h with 1 μ g/ml doxycycline prior to harvesting. Cells were then UV crosslinked, essentially as described above for the iCLIP procedure. Cell pellets were lysed in 2.2 mL iCLIP lysis buffer (50 mM Tris-HCl pH 7.4, 100 mM NaCl, 1% NP-40, 0.1% SDS, 0.5% sodium deoxycholate) containing protease inhibitor cocktail (Sigma-Aldrich P8340), sonicated and treated with 2 μ l Turbo DNase and RNase I (Ambion) for 5 min at 37°C. Note that RNase I dilution was experimentally determined during iCLIP experiments to yield optimal RNA fragment sizes (50 to 300 nucleotides). Under these lysis

conditions, physical protein-protein interactions are expected to be disrupted. The cell lysate was centrifuged in a microcentrifuge at maximum speed for 30min at 4°C. Clarified lysates were incubated with 6μl anti-GFP antibody (Thermo Fisher G10362) overnight followed by a 4-hour incubation with 60μl Protein G Dynabeads in a cold room (4°C). The beads were then washed 6 times with iCLIP high-salt buffer (50 mM Tris-HCl pH 7.4, 1 M NaCl, 1 mM EDTA, 1% NP-40, 0.1% SDS, 0.5% sodium deoxycholate) for 5 min each time at 4°C. The beads were then washed three times with AP high-salt buffer (10mM Tris-HCl, pH7.9, 420mM NaCl, 0.1% NP-40) with 5 min end-to-end rotation in the cold room. Beads were washed twice with non-detergent buffer (10mM Tris-HCl, pH7.9, 420mM NaCl). The immunoprecipitated proteins were eluted with 0.5M ammonium hydroxide (NH₄OH). The samples were prepared for mass spectrometry analysis as described below.

Experimental design for mass spectrometry

At least two biological replicates for each bait protein were processed independently in parallel with negative controls in each batch of samples. HEK293 cells expressing GFP only were used as control. Extensive washes were performed between samples to minimize carry-over. The order of sample acquisition on the mass spectrometer was also reversed for the second replicate to avoid systematic bias.

Sample preparation and proteomic analysis

IP material from each bait eluted in ammonium hydroxide was lyophilized. Proteins for mass spectrometry were prepared using in-solution trypsin digestion as described.^{65,73} Briefly, protein pellets were resuspended in 44 μL of 50mM ammonium bicarbonate. The sample was reduced with 100mM TCEP-HCL, alkylated with 500mM iodoacetamide, and digested with 1μg of trypsin (Promega V5111) overnight at 37°C. ZipTip pipette tips (EMD Millipore) were used for desalting of the samples using standard procedures. The samples were analyzed with an LTQ-Orbitrap Velos mass spectrometer (Thermo Scientific). Raw MS data was analyzed with X! Tandem as described,⁷⁴ which provided spectral counts for individual proteins. Finally, the MS data were analyzed using SAINTexpress against control purifications.⁵⁶ The prey proteins with Bayesian false discovery rate (BFDR) ≤ 0.01 were considered significant. Gene ontology (GO) enrichment analysis was carried out using ShinyGO⁵² as well as clusterProfiler.⁵³ Protein-protein interaction network was built using cytoscape.⁵⁵

QUANTIFICATION AND STATISTICAL ANALYSIS

iCLIP-seq data analysis

iCLIP- and miCLIP-seq data analysis were performed as detailed in our previous reports.^{14,65} Briefly, 51-nt (or longer) iCLIP-seq and miCLIP-seq raw reads that consisted of 3 random positions, a 4-nt multiplexing barcode, and another 2 random positions, followed by the cDNA sequence, were initially de-duplicated based on the first 45 nt. Reads were de-multiplexed, and we removed the random positions, barcodes, and any 3'-bases matching Illumina adaptors. We discarded any reads shorter than 25nt. The remaining reads were trimmed to 35nt. Trimming was performed using Trimmomatic.⁵⁰ Reads were mapped to the human genome/transcriptome (Ensembl annotation of hg19) using Tophat⁵⁷ with default settings. To prevent false assignments of reads from repetitive regions, any reads with a mapping quality < 3 were removed from further analysis. We utilized our published raw iCLIP-seq data for SP1, certain cleavage and polyadenylation factors (i.e., CPSF5, 7, 100, 160 and FIP1) (GEO: GSE165772) and MKRN2 (GEO: GSE136399), that were generated alongside the other data reported here. ZBTB48 and FTO iCLIP-seq data have been submitted in a companion paper (GEO: GSE228608)⁵⁰ and are publicly available. Moreover, a detailed analysis of the RPRD1B iCLIP-seq data was reported in a companion paper.⁷⁵ Finally, ZNF684 iCLIP-seq data, that were also generated as a part of this large-scale study, has been submitted as a separate companion paper elsewhere (GEO: GSE244136).⁴⁶

Identification of crosslink-induced truncations (CITS)

To define RNA-binding sites at single-nucleotide resolution, we utilized the CLIP Tool Kit (CTK) method.⁶¹ Since our method enriches for CITS, we observed minimal CIMS in our data. We called CITS on CLIP replicates and individual SMI samples. CITS with p -value ≤ 0.01 were kept for further analysis. CITS found in more than 50% of the iCLIP genes were defined as crosslink hotspots and were unconditionally removed from downstream analyses. The remaining iCLIP CITS were further normalized using SMI CITS. To this end, we used underlying reads from the iCLIP CITS and SMI CITS to perform a differential analysis using R package DESeq2 with a liberal cut off of $p \leq 0.5$. miCLIP data was processed similarly to iCLIP-seq data, and putative m⁶A sites were identified as CITS with the DRACH motif. Following the identification of CITS, we plotted them along mRNA transcripts by generating a metagene plot with the R package GenomicPlot (URL: <https://github.com/shuye2009/GenomicPlot>), as well as using MetaPlotR.⁵¹ Peak distribution across genomic regions (analyzing 5' and 3'UTRs separately) was also calculated using 'GenomicPlot' and represented as a bar chart.

Overlap of ChIP and iCLIP peaks and simulations

ChIP-seq data were acquired from the ENCODE consortium project, using 'ENCODE transcription factor ChIP-seq data' (https://www.encodeproject.org/chip-seq/transcription_factor/), and our previous reports^{8,21} (accession numbers: GEO: GSE76496 and GSE52523). ChIP-seq reads were mapped to the human genome hg19 using Bowtie 2 with default settings.⁷⁶ After

removal of duplicate reads, peaks were called jointly on immunoprecipitated and input samples with MACS2 (version 2.1.2).⁶² To assess the overlap of iCLIP peaks with ChIP-seq summit peaks, we extended the ChIP-seq summit peaks by 100 nt on either side (50nt on each side) (or other window sizes as indicated in the figures), and the number of iCLIP peaks that fell within this region were identified. Additionally, local enrichment of the iCLIP CITS around the ChIP-seq summit peaks was represented as a heatmap. In this representation, the iCLIP CITS were resized to 20 nt and the CITS signal density was plotted in a window of 1kb on each side of the ChIP-seq summit peaks.

Additionally, to compare the spatial relationship of the ChIP and CLIP binding sites of C2H2 factors, we simulated the following datasets: First, we generated an artificial set of binding sites by merging the ChIP profiles of all factors and randomly drawing a subset of peaks. Then, we constructed three prototype profiles to compare it with: (1) A set with perfect overlap consisting of a randomly drawn subset of 5% of these binding sites; (2) A similarly sized set, but with the binding sites shifted randomly by a distance that follows a normal distribution with mean=0 and SD=500 nt; (3) A random set independently drawn from the merged ChIP peaks, representing a random sample of 'typical' ChIP binding sites ('Random').

icSHAPE / iCLIP data amalgamation and visualization pipeline

Raw icSHAPE RNAseq reads from fractionation icSHAPE experiments in HEK293 cells were acquired from accession (GEO: GSE117840) and processed using Python with the tools and parameters outlined in the icSHAPE-pipe protocol as described previously.⁷⁷ To overlay RNA binding data with icSHAPE RNA structure data, we devised a computational icSHAPE/iCLIP overlap pipeline that scans through iCLIP peak coordinates and local sequence features and cross references these genomic locations with nucleotide-resolution icSHAPE reactivity scores. This overlay allowed us to estimate the local nucleotide accessibility and infer structure (strandedness) of C2H2-ZNFs' bound RNAs. To perform the overlap and comparison of icSHAPE RNA structural data with iCLIP-seq RNA binding results, custom scripts for each stage were written with the R statistical programming language. Scripts are freely available through Github (https://github.com/jburnsuoft/icSHAPE_iCLIP).

Sequence-specific icSHAPE analogue RNAplfold

To generate individual-nucleotide paired probability values from input sequences using a minimum free energy (MFE) sequence-based *in silico* methodology, we used a modified version of ViennaRNA's RNAplfold tool.²⁷ Sequences which were previously determined through iCLIP and icSHAPE overlap for each protein were used as input. The RNAplfold tool assigns a paired-probability score between 0 and 1 to each nucleotide along each input sequence, with 0 representing a 100% probability of a nucleotide being paired and 1 representing a 0% probability of being paired. The settings used were as follows: -winsize = 70, U -1 and -auto-id.

RNA structural prediction using *in silico* folding of C2H2-ZNF target sequences

For *in silico* folding of C2H2-ZNF-bound RNA sequences, ViennaRNA's RNAfold tool²⁷ was used as the primary folding algorithm for generating dot-bracket representations of bound RNA sequences for each protein. When incorporating SHAPE reactivity values into the folding algorithm, RNAfold requires both input sequence files in fasta format and a corresponding reactivity file (.shape), where each row contains the SHAPE reactivity score for its sequentially respective nucleotide in the fasta file. RNAfold was run with the following settings: -D2: which tells the algorithm to use a "dangling end" model, which adds the dangling energies for helix-adjacent bases to both sides of the helix; and -noLP: which prevents pairs that can only occur when isolated. To convert and incorporate SHAPE data, we used the setting shapemethod:"Z". With this setting, SHAPE reactivity scores are converted to pseudo energies through an equation derived from Zarringhalam et al. 2012,⁷⁸ which converts SHAPE reactivity into nucleotide paired probabilities with linear mapping. Two separate outputs were generated in dot-bracket format and visualized, one which selects the optimal folded conformation through MFE calculations, and the other selecting a centroid representative from thermodynamic ensemble secondary structures. Only MFE models were selected for downstream analysis.

The BEAM tool⁶³ employs a hidden Markov model approach to detect enrichment of structural motifs along input sequences that have been folded previously with RNAfold.²⁷ RNA sequences to be scanned with BEAM were provided in a multi-fasta format as the direct output from RNAfold runs. Each input sequence in the fasta file is represented by Three rows of information: the top row lists the MFE value for the folded sequence, followed by the chromosomal coordinates and strand information for the iCLIP-derived peak summit at the center of the sequence; the second row lists the RNA sequence; and the third row displays the predicted RNA secondary structure in dot-bracket format.

The BEAM tool (V.1.6.1)⁶³ was downloaded from (<http://beam.uniroma2.it/download>) and run for each protein with the following settings: -f: Multifasta file containing sequences and dot-bracket notation -w: Minimum motif width -W: Maximum motif width -C: "Clean mode" (1-3): 1. Exclude negative partials 2. Exclude partials under the 50% of the partial mean 3. Exclude partials under 90% of the mean. -M: Number of motifs to find (cutoff) -s: Total number of annealing steps -S: Total number of steps to perform -T: Starting temperature -r: Cooling rate -R: Random starts -n: Model Limit -k: Keep all runs, otherwise show best run -N: Search for primary sequence motifs too.

BEAM's background generation algorithm employs RNA sequences taken from the Rfam database, which considers sequences of similar length and fraction of paired nucleotides. BEAM's auto-generated background was used to establish adjusted P-values.

RNA-seq analysis

For differential mRNA expression analysis, reads were aligned to human genome GRCh38 using STAR (version 2.7.6.a),⁵⁸ and RSEM (version 1.3.3) was used to quantify gene-level read counts, which were then normalized by variance-stabilizing transformation using DESeq2.⁵⁴ Adjusted p-value ≤ 0.05 and absolute $\log_2$ fold change ± 1 were used as criteria to determine significantly regulated genes.

Alternative splicing analysis

Alternative splicing analysis was performed using vast-tools v2.2.2 in combination with VastDB version Hs2, released on Dec. 12, 2019, utilizing option-IR_version 1, essentially as described previously.⁶⁴ Changes greater than 10 dPSI/dPIR were considered significant and the expected minimum change was different from zero at $p > 0.95$ according to vast-tools' diff module. Events were filtered requiring a minimum of 10 reads per event and a balance score (quality score 4) of 'OK', 'B1' f, 'B2', 'BI' or 'Bn' for alternative exons or > 0.05 for intron retention events in at least 2 of 3 replicates. Overexpression RNA seq data for C2H2-ZNFs and control GFP-only were acquired from GEO: GSE76496.